

RNA-Guided Human Genome Engineering via Cas9

Prashant Mali,^{1,5} Luhan Yang,^{1,3,5} Kevin M. Esvelt,² John Aach,¹ Marc Guell,¹ James E. DiCarlo,⁴ Julie E. Norville,¹ George M. Church^{1,2*}

¹Department of Genetics, Harvard Medical School, Boston, MA 02115, USA. ²Wyss Institute for Biologically Inspired Engineering, Harvard University, Cambridge, MA 02138, USA. ³Biological and Biomedical Sciences Program, Harvard Medical School, Boston, MA 02115, USA. ⁴Department of Biomedical Engineering, Boston University, Boston, MA 02215, USA.

⁵These authors contributed equally to this work.

*To whom correspondence should be addressed. E-mail: gchurch@genetics.med.harvard.edu

Bacteria and archaea have evolved adaptive immune defenses termed clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated (Cas) systems that use short RNA to direct degradation of foreign nucleic acids. Here, we engineer the type II bacterial CRISPR system to function with custom guide RNA (gRNA) in human cells. For the endogenous AAVS1 locus, we obtained targeting rates of 10 to 25% in 293T cells, 13 to 8% in K562 cells, and 2 to 4% in induced pluripotent stem cells. We show this process relies on CRISPR components, is sequence-specific, and upon simultaneous introduction of multiple gRNAs, can effect multiplex editing of target loci. We also compute a genome-wide resource of ~190k unique gRNAs targeting ~40.5% of human exons. Our results establish an RNA-guided editing tool for facile, robust, and multiplexable human genome engineering.

Bacterial and archaeal CRISPR systems rely on crRNAs in complex with Cas proteins to direct degradation of complementary sequences present within invading viral and plasmid DNA (1–3). A recent *in vitro* reconstitution of the *S. pyogenes* type II CRISPR system demonstrated that crRNA fused to a normally trans-encoded tracrRNA is sufficient to direct Cas9 protein to sequence-specifically cleave target DNA sequences matching the crRNA (4). The fully defined nature of this 2-component system suggested that it might function in the cells of eukaryotic organisms such as yeast, plants, and even mammals. By cleaving genomic sequences targeted by RNA sequences (4–6), such a system could greatly enhance the ease of genome engineering.

Here we engineer the protein and RNA components of this bacterial type II CRISPR system in human cells. We began by synthesizing a human codon-optimized version of the Cas9 protein bearing a C terminus SV40 nuclear localization signal and cloning it into a mammalian expression system (Fig. 1A and fig. S1A). To direct Cas9 to cleave sequences of interest, we expressed crRNA-tracrRNA fusion transcripts, hereafter referred to as guide RNAs (gRNAs), from the human U6 polymerase III promoter. Importantly, directly transcribing gRNAs allowed us to avoid reconstituting the RNA processing machinery employed by bacterial CRISPR systems (Fig. 1A and fig. S1B) (4, 7–9). Constrained only by U6 transcription initiating with G and the requirement for the PAM (protospacer-adjacent motif) sequence -NGG following the 20 bp crRNA target, our highly versatile approach can in principle target any genomic site of the form GN₂₀GG (fig. S1C; see SI for a detailed discussion).

To test the functionality of our implementation for genome engineering, we developed a GFP reporter assay (Fig. 1B) in 293T cells similar to one previously described (10). Specifically, we established a stable cell line bearing a genomically integrated GFP coding sequence disrupted by the insertion of a stop codon and a 68bp genomic fragment from the AAVS1 locus that renders the expressed protein fragment non-

fluorescent. Homologous recombination (HR) using an appropriate repair donor can restore the normal GFP sequence, enabling us to quantify the resulting GFP⁺ cells by flow activated cell sorting (FACS).

To test the efficiency of our system at stimulating HR, we constructed two gRNAs, T1 and T2, that target the intervening AAVS1 fragment (Fig. 1B) and compared their activity to that of a previously described TAL effector nuclease heterodimer (TALEN) targeting the same region (11). We observed successful HR events using all three targeting reagents, with gene correction rates using the T1 and T2 gRNAs approaching 3% and 8% respectively (Fig. 1C). This RNA-mediated editing process was notably rapid, with the first detectable GFP⁺ cells appearing ~20 hours post transfection compared to ~40 hours for the AAVS1 TALENs. We observed HR only upon simultaneous introduction of the repair donor, Cas9 protein, and gRNA, confirming that all components are required for genome editing (fig. S2). While we noted no apparent toxicity associated with Cas9/crRNA expression, work with ZFNs and TALENs has shown

that nicking only one strand further reduces toxicity. Accordingly, we also tested a Cas9D10A mutant that known to function as a nickase *in vitro*, which yielded similar HR but lower non-homologous end joining (NHEJ) rates (fig. S3) (4, 5). Consistent with (4) where a related Cas9 protein is shown to cut both strands 6 bp upstream of the PAM, our NHEJ data confirmed that most deletions or insertions occurred at the 3' end of the target sequence (fig. S3B). We also confirmed that mutating the target genomic site prevents the gRNA from effecting HR at that locus, demonstrating that CRISPR-mediated genome editing is sequence specific (fig. S4). Finally, we showed that two gRNAs targeting sites in the GFP gene, and also three additional gRNAs targeting fragments from homologous regions of the DNA methyl transferase 3a (DNMT3a) and DNMT3b genes could sequence specifically induce significant HR in the engineered reporter cell lines (figs. S5 and S6). Together these results confirm that RNA-guided genome targeting in human cells is simple to execute and induces robust HR across multiple target sites.

Having successfully targeted an integrated reporter, we next turned to modifying a native locus. We used the gRNAs described above to target the AAVS1 locus located in the PPP1R12C gene on chromosome 19, which is ubiquitously expressed across most tissues (Fig. 2A) in 293Ts, K562s, and PGP1 human iPS cells (12) and analyzed the results by next-generation sequencing of the targeted locus. Consistent with our results for the GFP reporter assay, we observed high numbers of NHEJ events at the endogenous locus for all three cell types. The two gRNAs T1 and T2 achieved NHEJ rates of 10 and 25% in 293Ts, 13 and 38% in K562s, and 2 and 4% in PGP1-iPS cells, respectively (Fig. 2B). We observed no overt toxicity from the Cas9 and crRNA expression required to induce NHEJ in any of these cell types (fig. S7). As expected, NHEJ-mediated deletions for T1 and T2 were centered around the target site positions, further validating the sequence specificity of this targeting process (figs. S7 to S9). Interestingly, simultaneous introduction of both T1 and T2 gRNAs resulted in high efficiency deletion of the intervening

19bp fragment (fig. S8), demonstrating that multiplexed editing of genomic loci is feasible using this approach.

Lastly, we attempted to use HR to integrate either a dsDNA donor construct (13) or an oligo donor into the native AAVS1 locus (Fig. 2C and fig. S10). We confirmed HR-mediated integration using both approaches by PCR (Fig. 2D and fig. S10) and Sanger sequencing (Fig. 2E). We also readily derived 293T or iPS clones from the pool of modified cells using puromycin selection over two weeks (Fig. 2F and fig. S10). These results demonstrate that Cas9 is capable of efficiently integrating foreign DNA at endogenous loci in human cells.

Our versatile RNA-guided genome editing system can be readily adapted to modify other genomic sites by simply modifying the sequence of our gRNA expression vector to match a compatible sequence in the locus of interest. To facilitate this process, we bioinformatically generated ~190,000 specifically gRNA-targetable sequences targeting ~40.5% exons of genes in the human genome (refer to Methods and table S1). We incorporated these target sequences into a 200bp format compatible with multiplex synthesis on DNA arrays (14) (fig. S11 and tables S2 and S3). This resource provides a ready genome-wide reference of potential target sites in the human genome and a methodology for multiplex gRNA synthesis.

Our results demonstrate the promise of CRISPR-mediated gene targeting for RNA guided, robust and multiplexable mammalian genome engineering. The ease of retargeting our system to modify genomic sequences greatly exceeds that of comparable zinc finger nucleases (ZFNs) and TALENs while offering similar or greater efficiencies (4). Existing studies of Type II CRISPR specificity (4) suggest that target sites must perfectly match the PAM sequence NGG and the 8-12 base “seed sequence” at the 3’ end of the gRNA. The importance of the remaining 8-12 bases is less well understood and may depend on the binding strength of the matching gRNAs or on the inherent tolerance of Cas9 itself. Indeed, Cas9 will tolerate single mismatches at the 5’ end in bacteria and in vitro, suggesting that the 5’ G is not required. Moreover, it is likely the target locus’s underlying chromatin structure and epigenetic state will also impact the efficiency of genome editing in eukaryotic cells (13), although we suspect that Cas9’s helicase activity may render it more robust to these factors: but this remains to be evaluated. Elucidating the frequency and underlying causes of off-target nuclease activity (15, 16) induced by CRISPR, ZFN (17, 18) and TALEN (19, 20) genome engineering tools will be of utmost importance for safe genome modification and perhaps gene therapy. Potential avenues for improving CRISPR specificity include evaluating Cas9 homologs identified through bioinformatics and directed evolution of these nucleases toward higher specificity. Similarly, the range of CRISPR-targetable sequences could be expanded through the use of homologs with different PAM requirements (9), or directed evolution. Finally, inactivating one of the Cas9 nuclease domains increases the ratio of HR to NHEJ and may reduce toxicity (figs. S1A and fig. S3) (4, 5), while inactivating both domains may enable Cas9 to function as a retargetable DNA binding protein. As we explore these areas, we note that another parallel study (21) has independently confirmed the high efficiency of CRISPR-mediated gene targeting in mammalian cell lines. We expect RNA-guided genome targeting to have broad implications for synthetic biology (22, 23), the direct and multiplexed perturbation of gene networks (13, 24), and targeted ex vivo (25–27) and in vivo gene therapy (28).

References and Notes

1. B. Wiedenheft, S. H. Sternberg, J. A. Doudna, RNA-guided genetic silencing systems in bacteria and archaea. *Nature* **482**, 331 (2012). [doi:10.1038/nature10886](https://doi.org/10.1038/nature10886) [Medline](#)
2. D. Bhaya, M. Davison, R. Barrangou, CRISPR-Cas systems in bacteria and archaea: Versatile small RNAs for adaptive defense and regulation. *Annu. Rev. Genet.* **45**, 273 (2011). [doi:10.1146/annurev-genet-110410-132430](https://doi.org/10.1146/annurev-genet-110410-132430)

- [Medline](#)
3. M. P. Terns, R. M. Terns, CRISPR-based adaptive immune systems. *Curr. Opin. Microbiol.* **14**, 321 (2011). [doi:10.1016/j.mib.2011.03.005](https://doi.org/10.1016/j.mib.2011.03.005) [Medline](#)
4. M. Jinek *et al.*, A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science* **337**, 816 (2012). [doi:10.1126/science.1225829](https://doi.org/10.1126/science.1225829) [Medline](#)
5. G. Gasiunas, R. Barrangou, P. Horvath, V. Siksnys, Cas9-crRNA ribonucleoprotein complex mediates specific DNA cleavage for adaptive immunity in bacteria. *Proc. Natl. Acad. Sci. U.S.A.* **109**, E2579 (2012). [doi:10.1073/pnas.1208507109](https://doi.org/10.1073/pnas.1208507109) [Medline](#)
6. R. Sapranasuskas *et al.*, The *Streptococcus thermophilus* CRISPR/Cas system provides immunity in *Escherichia coli*. *Nucleic Acids Res.* **39**, 9275 (2011). [doi:10.1093/nar/gkr606](https://doi.org/10.1093/nar/gkr606) [Medline](#)
7. T. R. Brummelkamp, R. Bernards, R. Agami, A system for stable expression of short interfering RNAs in mammalian cells. *Science* **296**, 550 (2002). [doi:10.1126/science.1068999](https://doi.org/10.1126/science.1068999) [Medline](#)
8. M. Miyagishi, K. Taira, U6 promoter-driven siRNAs with four uridine 3’ overhangs efficiently suppress targeted gene expression in mammalian cells. *Nat. Biotechnol.* **20**, 497 (2002). [doi:10.1038/nbt0502-497](https://doi.org/10.1038/nbt0502-497) [Medline](#)
9. E. Deltcheva *et al.*, CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III. *Nature* **471**, 602 (2011). [doi:10.1038/nature09886](https://doi.org/10.1038/nature09886) [Medline](#)
10. J. Zou, P. Mali, X. Huang, S. N. Dowey, L. Cheng, Site-specific gene correction of a point mutation in human iPS cells derived from an adult patient with sickle cell disease. *Blood* **118**, 4599 (2011). [doi:10.1182/blood-2011-02-335554](https://doi.org/10.1182/blood-2011-02-335554) [Medline](#)
11. N. E. Sanjana *et al.*, A transcription activator-like effector toolbox for genome engineering. *Nat. Protoc.* **7**, 171 (2012). [doi:10.1038/nprot.2011.431](https://doi.org/10.1038/nprot.2011.431) [Medline](#)
12. J. H. Lee *et al.*, A robust approach to identifying tissue-specific gene expression regulatory variants using personalized human induced pluripotent stem cells. *PLoS Genet.* **5**, e1000718 (2009). [doi:10.1371/journal.pgen.1000718](https://doi.org/10.1371/journal.pgen.1000718) [Medline](#)
13. D. Hockemeyer *et al.*, Efficient targeting of expressed and silent genes in human ESCs and iPSCs using zinc-finger nucleases. *Nat. Biotechnol.* **27**, 851 (2009). [doi:10.1038/nbt.1562](https://doi.org/10.1038/nbt.1562) [Medline](#)
14. S. Kosuri *et al.*, Scalable gene synthesis by selective amplification of DNA pools from high-fidelity microchips. *Nat. Biotechnol.* **28**, 1295 (2010). [doi:10.1038/nbt.1716](https://doi.org/10.1038/nbt.1716) [Medline](#)
15. V. Pattanayak, C. L. Ramirez, J. K. Joung, D. R. Liu, Revealing off-target cleavage specificities of zinc-finger nucleases by in vitro selection. *Nat. Methods* **8**, 765 (2011). [doi:10.1038/nmeth.1670](https://doi.org/10.1038/nmeth.1670) [Medline](#)
16. N. M. King, O. Cohen-Haguenu, En route to ethical recommendations for gene transfer clinical trials. *Mol. Ther.* **16**, 432 (2008). [doi:10.1038/mt.2008.13](https://doi.org/10.1038/mt.2008.13) [Medline](#)
17. Y. G. Kim, J. Cha, S. Chandrasegaran, Hybrid restriction enzymes: Zinc finger fusions to Fok I cleavage domain. *Proc. Natl. Acad. Sci. U.S.A.* **93**, 1156 (1996). [doi:10.1073/pnas.93.3.1156](https://doi.org/10.1073/pnas.93.3.1156) [Medline](#)
18. E. J. Rebar, C. O. Pabo, Zinc finger phage: Affinity selection of fingers with new DNA-binding specificities. *Science* **263**, 671 (1994). [doi:10.1126/science.8303274](https://doi.org/10.1126/science.8303274) [Medline](#)
19. J. Boch *et al.*, Breaking the code of DNA binding specificity of TAL-type III effectors. *Science* **326**, 1509 (2009). [doi:10.1126/science.1178811](https://doi.org/10.1126/science.1178811) [Medline](#)
20. M. J. Moscou, A. J. Bogdanove, A simple cipher governs DNA recognition by TAL effectors. *Science* **326**, 1501 (2009). [doi:10.1126/science.1178817](https://doi.org/10.1126/science.1178817) [Medline](#)
21. L. Cong *et al.*, Multiplex genome engineering using CRISPR/Cas systems. *Science* **10.1126/science.1231143** (2013).
22. A. S. Khalil, J. J. Collins, Synthetic biology: Applications come of age. *Nat. Rev. Genet.* **11**, 367 (2010). [doi:10.1038/nrg2775](https://doi.org/10.1038/nrg2775) [Medline](#)
23. P. E. Purnick, R. Weiss, The second wave of synthetic biology: From modules to systems. *Nat. Rev. Mol. Cell Biol.* **10**, 410 (2009). [doi:10.1038/nrm2698](https://doi.org/10.1038/nrm2698) [Medline](#)
24. J. Zou *et al.*, Gene targeting of a disease-related gene in human induced pluripotent stem and embryonic stem cells. *Cell Stem Cell* **5**, 97 (2009). [doi:10.1016/j.stem.2009.05.023](https://doi.org/10.1016/j.stem.2009.05.023) [Medline](#)
25. N. Holt *et al.*, Human hematopoietic stem/progenitor cells modified by zinc-finger nucleases targeted to CCR5 control HIV-1 in vivo. *Nat. Biotechnol.* **28**, 839 (2010). [doi:10.1038/nbt.1663](https://doi.org/10.1038/nbt.1663) [Medline](#)
26. F. D. Urnov *et al.*, Highly efficient endogenous human gene correction using

- designed zinc-finger nucleases. *Nature* **435**, 646 (2005). [doi:10.1038/nature03556](https://doi.org/10.1038/nature03556) [Medline](#)
27. A. Lombardo *et al.*, Gene editing in human stem cells using zinc finger nucleases and integrase-defective lentiviral vector delivery. *Nat. Biotechnol.* **25**, 1298 (2007). [doi:10.1038/nbt1353](https://doi.org/10.1038/nbt1353) [Medline](#)
 28. H. Li *et al.*, In vivo genome editing restores haemostasis in a mouse model of haemophilia. *Nature* **475**, 217 (2011). [doi:10.1038/nature10177](https://doi.org/10.1038/nature10177) [Medline](#)
 29. K. S. Makarova *et al.*, Evolution and classification of the CRISPR-Cas systems. *Nat. Rev. Microbiol.* **9**, 467 (2011). [doi:10.1038/nrmicro2577](https://doi.org/10.1038/nrmicro2577) [Medline](#)
 30. P. Horvath, R. Barrangou, CRISPR/Cas, the immune system of bacteria and archaea. *Science* **327**, 167 (2010). [doi:10.1126/science.1179555](https://doi.org/10.1126/science.1179555) [Medline](#)
 31. H. Deveau *et al.*, Phage response to CRISPR-encoded resistance in *Streptococcus thermophilus*. *J. Bacteriol.* **190**, 1390 (2008). [doi:10.1128/JB.01412-07](https://doi.org/10.1128/JB.01412-07) [Medline](#)
 32. J. R. van der Ploeg, Analysis of CRISPR in *Streptococcus mutans* suggests frequent occurrence of acquired immunity against infection by M102-like bacteriophages. *Microbiology* **155**, 1966 (2009). [doi:10.1099/mic.0.027508-0](https://doi.org/10.1099/mic.0.027508-0) [Medline](#)
 33. M. Rho, Y. W. Wu, H. Tang, T. G. Doak, Y. Ye, Diverse CRISPRs evolving in human microbiomes. *PLoS Genet.* **8**, e1002441 (2012). [doi:10.1371/journal.pgen.1002441](https://doi.org/10.1371/journal.pgen.1002441) [Medline](#)
 34. D. T. Pride *et al.*, Analysis of streptococcal CRISPRs from human saliva reveals substantial sequence diversity within and between subjects over time. *Genome Res.* **21**, 126 (2011). [doi:10.1101/gr.111732.110](https://doi.org/10.1101/gr.111732.110) [Medline](#)
 35. J. E. Garneau *et al.*, The CRISPR/Cas bacterial immune system cleaves bacteriophage and plasmid DNA. *Nature* **468**, 67 (2010). [doi:10.1038/nature09523](https://doi.org/10.1038/nature09523) [Medline](#)
 36. K. M. Esvelt, J. C. Carlson, D. R. Liu, A system for the continuous directed evolution of biomolecules. *Nature* **472**, 499 (2011). [doi:10.1038/nature09929](https://doi.org/10.1038/nature09929) [Medline](#)
 37. R. Barrangou, P. Horvath, CRISPR: New horizons in phage resistance and strain identification. *Annu. Rev. Food Sci. Technol.* **3**, 143 (2012). [doi:10.1146/annurev-food-022811-101134](https://doi.org/10.1146/annurev-food-022811-101134) [Medline](#)
 38. W. J. Kent *et al.*, The human genome browser at UCSC. *Genome Res.* **12**, 996 (2002). [Medline](#)
 39. T. R. Dreszer *et al.*, The UCSC Genome Browser database: Extensions and updates 2011. *Nucleic Acids Res.* **40**, (Database issue), D918 (2012). [doi:10.1093/nar/gkr1055](https://doi.org/10.1093/nar/gkr1055) [Medline](#)
 40. D. Karolchik *et al.*, The UCSC Table Browser data retrieval tool. *Nucleic Acids Res.* **32**, (Database issue), D493 (2004). [doi:10.1093/nar/gkh103](https://doi.org/10.1093/nar/gkh103) [Medline](#)
 41. A. R. Quinlan, I. M. Hall, BEDTools: A flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841 (2010). [doi:10.1093/bioinformatics/btq033](https://doi.org/10.1093/bioinformatics/btq033) [Medline](#)
 42. B. Langmead, C. Trapnell, M. Pop, S. L. Salzberg, Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol.* **10**, R25 (2009). [doi:10.1186/gb-2009-10-3-r25](https://doi.org/10.1186/gb-2009-10-3-r25) [Medline](#)
 43. R. Lorenz *et al.*, ViennaRNA Package 2.0. *Algorithms Mol. Biol.* **6**, 26 (2011). [doi:10.1186/1748-7188-6-26](https://doi.org/10.1186/1748-7188-6-26) [Medline](#)
 44. D. H. Mathews, J. Sabina, M. Zuker, D. H. Turner, Expanded sequence dependence of thermodynamic parameters improves prediction of RNA secondary structure. *J. Mol. Biol.* **288**, 911 (1999). [doi:10.1006/jmbi.1999.2700](https://doi.org/10.1006/jmbi.1999.2700) [Medline](#)
 45. R. E. Thurman *et al.*, The accessible chromatin landscape of the human genome. *Nature* **489**, 75 (2012). [doi:10.1038/nature11232](https://doi.org/10.1038/nature11232) [Medline](#)
 46. Q. Xu, M. R. Schlach, G. J. Hannon, S. J. Elledge, Design of 240,000 orthogonal 25mer DNA barcode probes. *Proc. Natl. Acad. Sci. U.S.A.* **106**, 2289 (2009). [doi:10.1073/pnas.0812506106](https://doi.org/10.1073/pnas.0812506106) [Medline](#)

Acknowledgments: This work was supported by NIH grant P50 HG005550. We thank Sriram Kosuri for advice on the oligonucleotide pool designs and synthesis. G.M.C. and P.M. have filed a patent based on the findings of this study.

Supplementary Materials

www.sciencemag.org/cgi/content/full/science.1232033/DC1

Materials and Methods

Figs. S1 to S11

Tables S1 to S3

References (29–46)

26 October 2012; accepted 12 December 2012

Published online 03 January 2013; 10.1126/science.1232033

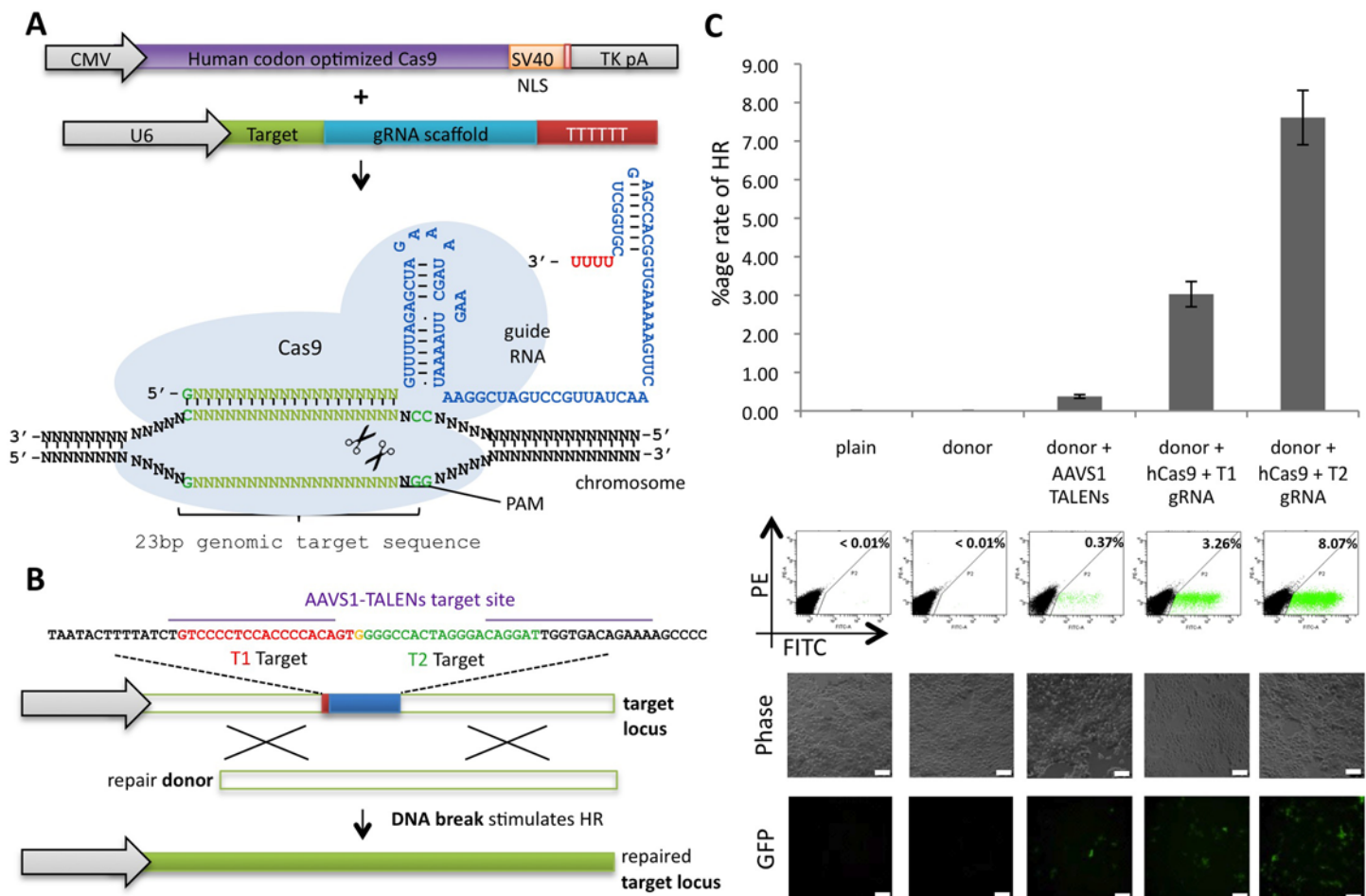


Fig. 1. Genome editing in human cells using an engineered type II CRISPR system. **(A)** RNA-guided gene targeting in human cells involves co-expression of the Cas9 protein bearing a C terminus SV40 nuclear localization signal with one or more guide RNAs (gRNAs) expressed from the human U6 polymerase III promoter. Cas9 unwinds the DNA duplex and cleaves both strands upon recognition of a target sequence by the gRNA, but only if the correct protospacer-adjacent motif (PAM) is present at the 3' end. Any genomic sequence of the form GN₂₀GG can in principle be targeted. **(B)** A genomically integrated GFP coding sequence is disrupted by the insertion of a stop codon and a 68bp genomic fragment from the AAVS1 locus. Restoration of the GFP sequence by homologous recombination (HR) with an appropriate donor sequence results in GFP⁺ cells that can be quantitated by FACS. T1 and T2 gRNAs target sequences within the AAVS1 fragment. Binding sites for the two halves of the TAL effector nuclease heterodimer (TALEN) are underlined. **(C)** Bar graph depicting HR efficiencies induced by T1, T2, and TALEN-mediated nuclease activity at the target locus, as measured by FACS. Representative FACS plots and microscopy images of the targeted cells are depicted below (scale bar is 100 microns). Data are means $\pm$ SEM (N=3).

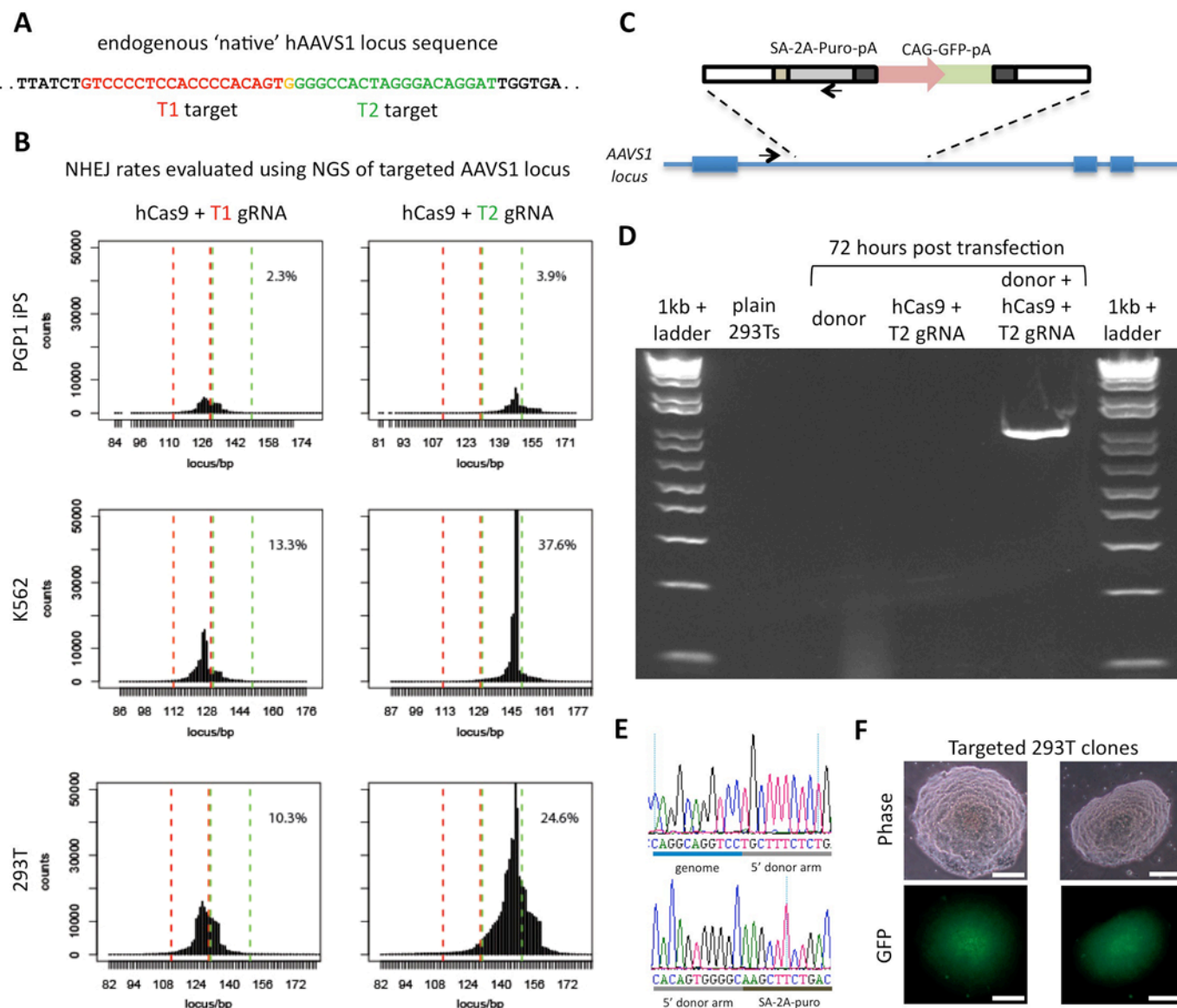


Fig. 2. RNA-guided genome editing of the native AAVS1 locus in multiple cell types. **(A)** T1 (red) and T2 (green) gRNAs target sequences in an intron of the PPP1R12C gene within the chromosome 19 AAVS1 locus. **(B)** Total count and location of deletions caused by NHEJ in 293Ts, K562s, and PGP1 iPS cells following expression of Cas9 and either T1 or T2 gRNAs as quantified by next-generation sequencing. Red and green dash lines demarcate the boundaries of the T1 and T2 gRNA targeting sites. NHEJ frequencies for T1 and T2 gRNAs were 10% and 25% in 293T, 13% and 38% in K562, and 2% and 4% in PGP1 iPS cells, respectively. **(C)** DNA donor architecture for HR at the AAVS1 locus, and the locations of sequencing primers (arrows) for detecting successful targeted events, are depicted. **(D)** PCR assay three days post transfection demonstrates that only cells expressing the donor, Cas9 and T2 gRNA exhibit successful HR events. **(E)** Successful HR was confirmed by Sanger sequencing of the PCR amplicon showing that the expected DNA bases at both the genome-donor and donor-insert boundaries are present. **(F)** Successfully targeted clones of 293T cells were selected with puromycin for 2 weeks. Microscope images of two representative GFP⁺ clones is shown (scale bar is 100 microns).



Supplementary Materials for **RNA-Guided Human Genome Engineering via Cas9**

Prashant Mali, Luhan Yang, Kevin M. Esvelt, John Aach, Marc Guell, James E. DiCarlo,
Julie E. Norville, George M. Church*

*To whom correspondence should be addressed. E-mail: gchurch@genetics.med.harvard.edu

Published 3 January 2013 on *Science Express*
DOI: 10.1126/science.1232033

This PDF file includes

Materials and Methods
Supplementary Text
Figs. S1 to S11
Full References

Other Supplementary Material for this manuscript includes the following:

(available at http://arep.med.harvard.edu/human_crispr)

Table S1. Bioinformatically computed genome-wide resource of candidate unique gRNA targets in human exons.

Table S2. Incorporation of gRNA targets in Table 1 into a 200-bp format suitable for multiplex DNA array based synthesis.

Table S3. 12k gRNA targets from Table 1 in a 200-bp format synthesized by CustomArray Inc.

Table of Contents

Section	Page
Supplementary Text: The Type II CRISPR-Cas System	3
Material and Methods	7
Fig. S1. The engineered type II CRISPR system for human cells.	13
Fig. S2. RNA-guided genome editing requires both Cas9 and guide RNA for successful targeting.	15
Fig. S3. Analysis of gRNA and Cas9 mediated genome editing.	16
Fig. S4. RNA-guided genome editing is target sequence specific.	18
Fig. S5. Guide RNAs targeted to the GFP sequence enable robust genome editing.	20
Fig. S6. RNA-guided genome editing is target sequence specific, and demonstrates similar targeting efficiencies as ZFNs or TALENs.	21
Fig. S7. RNA-guided NHEJ in human iPS cells.	23
Fig. S8. RNA-guided NHEJ in human K562 cells.	25
Fig. S9. RNA-guided NHEJ in human 293T cells.	27
Fig. S10. HR at the endogenous AAVS1 locus using either a dsDNA donor or a short oligonucleotide donor.	29
Fig. S11. Methodology for multiplex synthesis, retrieval and U6 expression vector cloning of guide RNAs targeting genes in the human genome.	31
Table S1. Bioinformatically computed genome-wide resource of candidate unique gRNA targets in human exons.	*
Table S2. Incorporation of gRNA targets in Table 1 into a 200bp format suitable for multiplex DNA array based synthesis.	*
Table S3. 12k gRNA targets from Table 1 in a 200bp format synthesized by CustomArray Inc.	*

*Because of size constraints, the Tables S1, S2, and S3 are available on: http://arep.med.harvard.edu/human_crispr.

The Type II CRISPR-Cas System

Bacteria and archaea have evolved adaptive immune defenses termed clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated (Cas) systems that use short RNA to direct degradation of foreign nucleic acids. CRISPR defense involves acquisition and integration of new targeting “spacers” from invading virus or plasmid DNA into the CRISPR locus, expression and processing of short guiding CRISPR RNAs (crRNAs) consisting of spacer-repeat units, and cleavage of nucleic acids (most commonly DNA) complementary to the spacer.

Three classes of CRISPR systems have been described thus far (Type I, II and III). Here we focus on the Type II CRISPR system, which utilizes a single effector enzyme, Cas9, to cleave dsDNA, whereas Type I and Type III systems require multiple distinct effectors acting as a complex (for a detailed review of CRISPR classification, see reference (29)). As a consequence, Type II systems are more likely to function in alternative contexts such as eukaryotic cells. The Type II effector system consists of a long pre-crRNA transcribed from the spacer-containing CRISPR locus, the multifunctional Cas9 protein, and a tracrRNA important for gRNA processing. The tracrRNAs hybridize to the repeat regions separating the spacers of the pre-crRNA, initiating dsRNA cleavage by endogenous RNase III, which is followed by a second cleavage event within each spacer by Cas9, producing mature crRNAs that remain associated with the tracrRNA and Cas9. Jinek et al. demonstrated that a tracrRNA-crRNA fusion, termed a guide RNA (gRNA) in this work (Fig. 1), is functional *in vitro*, obviating the need for RNase III and the crRNA processing in general (4).

Type II CRISPR interference is a result of Cas9 unwinding the DNA duplex and searching for sequences matching the crRNA to cleave. Target recognition occurs upon detection of complementarity between a “protospacer” sequence in the target DNA and the remaining spacer sequence in the crRNA. Importantly, Cas9 cuts the DNA only if a correct protospacer-adjacent motif (PAM) is also present at the 3’ end. Different Type II systems have differing PAM requirements. The *S. pyogenes* system utilized in this work requires an NGG sequence, where N can be any nucleotide. *S. thermophilus* Type II systems require NGGNG (30) and NNAGAAW (31), respectively, while different *S. mutans* systems tolerate NGG or NAAR (32). Bioinformatic analyses have generated extensive databases of CRISPR loci in a variety of bacteria that may serve to identify new PAMs and expand the set of CRISPR-targetable sequences (33, 34). In *S. thermophilus*, Cas9 generates a blunt-ended double-stranded break 3bp prior to the 3’ end of the protospacer (5), a process mediated by two catalytic domains in the Cas9 protein: an HNH domain that cleaves the complementary strand of the DNA and a RuvC-like domain that cleaves the non-complementary strand (refer Fig. 1A, fig. S1). While the *S. pyogenes* system has not been characterized to the same level of precision, DSB formation also occurs towards the 3’ end of the protospacer. If one of the two nuclease domains is inactivated, Cas9 will function as a nickase *in vitro* (4) and in human cells (fig. S3).

As a genome engineering tool, the specificity of gRNA-directed Cas9 cleavage will be of the utmost importance. Significant off-target activity could cause unwanted double-strand breaks at other regions of the genome, resulting in toxicity and possibly oncogenesis in gene therapy applications. The *S. pyogenes* system tolerates mismatches in the first 6 bases out of the 20bp mature spacer sequence *in vitro*. However, it is entirely possible that greater

stringency is required *in vivo* given the low toxicity we observed in human cell lines, as potential off-target sites matching (last 14 bp) NGG exist within the human reference genome for our gRNAs. Mismatches towards the 3' end of the spacer, known as the “seed sequence” (6), are less well tolerated. Jinek et al. found that single mismatches in the PAM at positions -3 through -7 abolished interference *in vitro*, though a mismatch at position -10 did not (4). In *S. thermophilus*, single mutations in the PAM or at positions -1, -3 through -5, and -7 through -8 abolished interference. When transplanted into *E. coli*, the *S. thermophilus* system did not tolerate single mutations in the PAM or in positions -3, -6, or -8. As a caveat, Garneau et al. found that spacers acquired from plasmid DNA tolerated greater degeneracy in both the PAM and seed sequence while sufficing to block plasmid acquisition in *S. thermophilus* (35); however, similar degeneracy was not sufficient to block phage infection (31), emphasizing the importance of the assay utilized. Taken together, these results point towards the urgent need to assay specificity in the context of interest.

There exist at least four possible ways to improve specificity. First, it is possible that interference is sensitive to the melting temperature of the gRNA-DNA hybrid, in which case AT-rich target sequences are likely to have fewer off-target sites. Second, carefully choosing target sites to avoid pseudo-sites with at least 14bp matching sequences elsewhere in the genome of interest. Third, the use of a Cas9 variant requiring a longer PAM sequence will reduce the set of potential targetable sequences, but should similarly reduce the frequency of off-target sites. Finally, directed evolution might be utilized to improve Cas9 specificity to a level sufficient to completely preclude off-target activity, ideally requiring a perfect 20bp gRNA match with a minimal PAM. Such a project is likely to require extensive modifications to the Cas9 protein, as

the small genomes of bacteria are unlikely to select for high specificity in natural variants. As such, novel methods permitting many rounds of evolution in a short timeframe (36) may be warranted. For more detailed reviews of CRISPR systems, see references (1, 37).

Material and Methods

Plasmid construction

The Cas9 gene sequence was human codon optimized and assembled by hierarchical fusion PCR assembly of 9 500bp gBlocks ordered from IDT (sequence in fig. S1A). Cas9_D10A was similarly constructed. The resulting full-length products were cloned into the pcDNA3.3-TOPO vector (Invitrogen). The target gRNA expression constructs were directly ordered as individual 455bp gBlocks from IDT (sequence in fig. S1B) and either cloned into the pCR-BluntII-TOPO vector (Invitrogen) or pcr amplified. The vectors for the HR reporter assay involving a broken GFP were constructed by fusion PCR assembly of the GFP sequence bearing the stop codon and 68bp AAVS1 fragment (or mutants thereof; refer fig. S4), or 58bp fragments from the DNMT3a and DNMT3b genomic loci (refer fig. S6) assembled into the EGIP lentivector from Addgene (plasmid #26777). These lentivectors were then used to establish the GFP reporter stable lines. TALENs used in this study were constructed using the protocols described in (11). All DNA reagents developed in this study are available at Addgene.

Cell culture

PGP1 iPS cells were maintained on Matrigel (BD Biosciences)-coated plates in mTeSR1 (Stemcell Technologies). Cultures were passaged every 5–7 d with TrypLE Express (Invitrogen). K562 cells were grown and maintained in RPMI (Invitrogen) containing 15% FBS. HEK 293T cells were cultured in Dulbecco's modified Eagle's medium (DMEM, Invitrogen) high glucose supplemented with 10% fetal bovine serum (FBS, Invitrogen), penicillin/streptomycin (pen/strep, Invitrogen), and non-essential amino acids (NEAA, Invitrogen). All cells were maintained at 37°C and 5% CO₂ in a humidified incubator.

Gene targeting of PGP1 iPS, K562 and 293Ts

PGP1 iPS cells were cultured in Rho kinase (ROCK) inhibitor (Calbiochem) 2h before nucleofection. Cells were harvest using TrypLE Express (Invitrogen) and 2×10^6 cells were resuspended in P3 reagent (Lonza) with 1 μ g Cas9 plasmid, 1 μ g gRNA and/or 1 μ g DNA donor plasmid, and nucleofected according to manufacturer's instruction (Lonza). Cells were subsequently plated on an mTeSR1-coated plate in mTeSR1 medium supplemented with ROCK inhibitor for the first 24h. For K562s, 2×10^6 cells were resuspended in SF reagent (Lonza) with 1 μ g Cas9 plasmid, 1 μ g gRNA and/or 1 μ g DNA donor plasmid, and nucleofected according to manufacturer's instruction (Lonza). For 293Ts, 0.1×10^6 cells were transfected with 1 μ g Cas9 plasmid, 1 μ g gRNA and/or 1 μ g DNA donor plasmid using Lipofectamine 2000 as per the manufacturer's protocols. The DNA donors used for endogenous AAVS1 targeting were either a dsDNA donor (Fig. 2C) or a 90mer oligonucleotide. The former has flanking short homology arms and a SA-2A-puromycin-CaGGS-eGFP cassette to enrich for successfully targeted cells.

Assess the targeting efficiency

Cells were harvested 3 days after nucleofection and the genomic DNA of $\sim 1 \times 10^6$ cells was extracted using prepGEM (ZyGEM). PCR was conducted to amplify the targeting region with genomic DNA derived from the cells and amplicons were deep sequenced by MiSeq Personal Sequencer (Illumina) with coverage >200,000 reads. The sequencing data was analyzed to estimate NHEJ efficiencies. The reference AAVS1 sequence analyzed is:

```
CACTTCAGGACAGCATGTTTGCTGCCTCCAGGGATCCTGTGTCCCCGAGCTGGGACCACCTTATATTTCC  
AGGGCCGGTTAATGTGGCTCTGGTTCTGGGTACTTTTATCTGTCCCCTCCACCCCACAGTGGGGCCACTA
```


GGGACAGGATTGGTGACAGAAAAGCCCCATCCTTAGGCCTCCTCCTCCTAGTCTCCTGATATTGGGTCT
AACCCCCACCTCCTGTTAGGCAGATTCCTTATCTGGTGACACACCCCCATTCCTGGA

The PCR primers for amplifying the targeting regions in the human genome are:

AAVS1-R	CTCGGCATTCTGCTGAACCGCTCTTCCGATCTacaggaggtgggggtagac
AAVS1-F.1	ACACTCTTCCCTACACGACGCTCTTCCGATCTCGTGATtatattcccagggccgggta
AAVS1-F.2	ACACTCTTCCCTACACGACGCTCTTCCGATCTACATCGtatattcccagggccgggta
AAVS1-F.3	ACACTCTTCCCTACACGACGCTCTTCCGATCTGCCTAAtatattcccagggccgggta
AAVS1-F.4	ACACTCTTCCCTACACGACGCTCTTCCGATCTTGGTCAtatattcccagggccgggta
AAVS1-F.5	ACACTCTTCCCTACACGACGCTCTTCCGATCTCACTGTtatattcccagggccgggta
AAVS1-F.6	ACACTCTTCCCTACACGACGCTCTTCCGATCTATTGGCtatattcccagggccgggta
AAVS1-F.7	ACACTCTTCCCTACACGACGCTCTTCCGATCTGATCTGtatattcccagggccgggta
AAVS1-F.8	ACACTCTTCCCTACACGACGCTCTTCCGATCTTCAAGTtatattcccagggccgggta
AAVS1-F.9	ACACTCTTCCCTACACGACGCTCTTCCGATCTCTGATCtatattcccagggccgggta
AAVS1-F.10	ACACTCTTCCCTACACGACGCTCTTCCGATCTAAGCTAAtatattcccagggccgggta
AAVS1-F.11	ACACTCTTCCCTACACGACGCTCTTCCGATCTGTAGCCtatattcccagggccgggta
AAVS1-F.12	ACACTCTTCCCTACACGACGCTCTTCCGATCTTACAAGtatattcccagggccgggta

To analyze the HR events using the DNA donor in Fig. 2C the primers used were:

HR_AAVS1-F	CTGCCGTCTCTCTCCTGAGT
HR_Puro-R	GTGGGCTTGTA CTGGTCAT

Bioinformatics approach for computing human exon CRISPR targets and methodology for their multiplexed synthesis

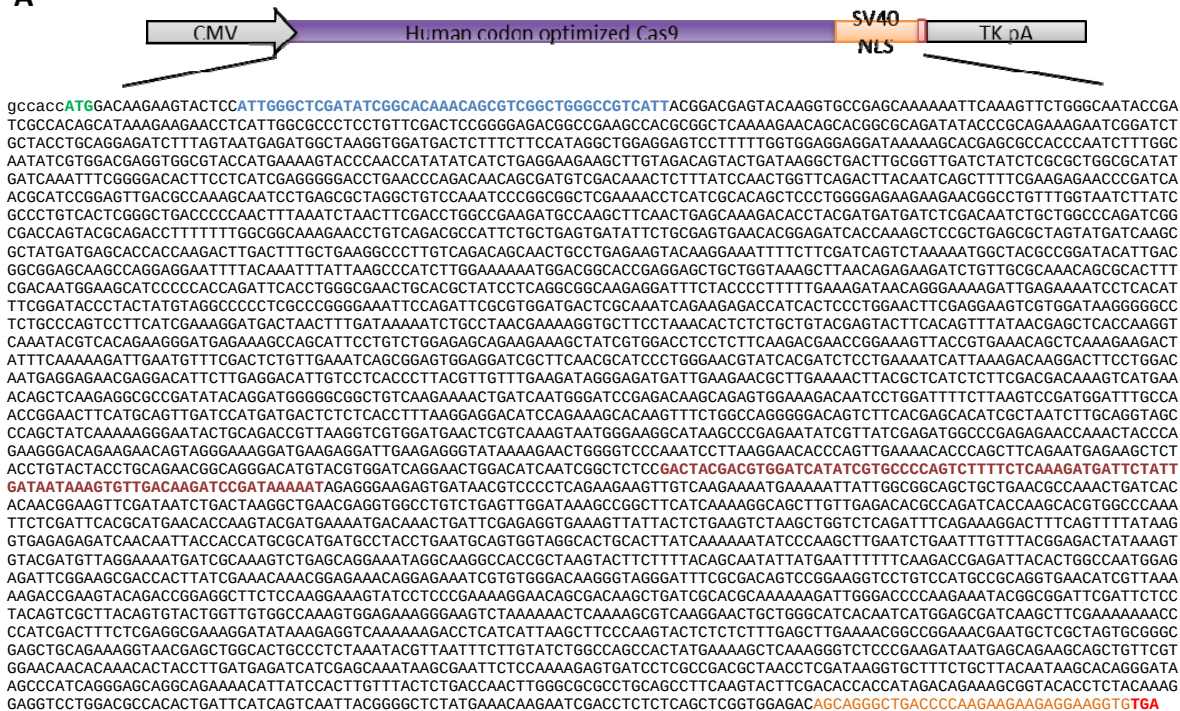
We sought to generate a set of gRNA gene sequences that maximally target specific locations in human exons but minimally target other locations in the genome. Maximally efficient targeting by a gRNA is achieved by 23nt sequences, the 5'-most 20nt of which exactly complement a desired location, while the three 3'-most bases must be of the form NGG. Additionally, the 5'-most nt must be a G to establish a pol-III transcription start site. However, according to (4), mispairing of the six 5'-most nt of a 20bp gRNA against its genomic target does not abrogate Cas9-mediated cleavage so long as the last 14nt pairs properly, but mispairing of the eight 5'-most nt along with pairing of the last 12 nt does, while the case of the seven 5'-most nt mispairs and 13 3' pairs was not tested. To be conservative regarding off-target effects, we therefore assumed that the case of the seven 5'-most mispairs is, like the case of six, permissive of cleavage, so that pairing of the 3'-most 13nt is sufficient for cleavage. To identify CRISPR target sites within human exons that should be cleavable without off-target cuts, we therefore examined all 23bp sequences of the form 5'-GBBBB BBBB BBBB BBBB NGG-3' (*form 1*), where the B's represent the bases at the exon location, for which no sequence of the form 5'-NNNNN NNBBB BBBB BBBB NGG-3' (*form 2*) existed at any other location in the human genome. Specifically, we (i) downloaded a BED file of locations of coding regions of all RefSeq genes the GRCh37/hg19 human genome from the UCSC Genome Browser (38-40). Coding exon locations in this BED file comprised a set of 346089 mappings of RefSeq mRNA accessions to the hg19 genome. However, some RefSeq mRNA accessions mapped to multiple genomic locations (probable gene duplications), and many accessions mapped to subsets of the same set of exon locations (multiple isoforms of the same genes). To distinguish apparently duplicated gene instances and consolidate multiple references to the same genomic exon instance by multiple

RefSeq isoform accessions, we therefore (ii) added unique numerical suffixes to 705 RefSeq accession numbers that had multiple genomic locations, and (iii) used the mergeBed function of BEDTools (41) (v2.16.2-zip-87e3926) to consolidate overlapping exon locations into merged exon regions. These steps reduced the initial set of 346089 RefSeq exon locations to 192783 distinct genomic regions. We then downloaded the hg19 sequence for all merged exon regions using the UCSC Table Browser, adding 20bp of padding on each end. (iv) Using custom perl code, we identified 1657793 instances of *form 1* within this exonic sequence. (v) We then filtered these sequences for the existence of off-target occurrences of *form 2*: For each merged exon *form 1* target, we extracted the 3'-most 13bp specific (B) "core" sequences and, for each core generated the four 16bp sequences 5'-BBB BBBB BBBB NGG-3' (N = A, C, G, and T), and searched the entire hg19 genome for exact matches to these 6631172 sequences using Bowtie version 0.12.8 (42) using the parameters -l 16 -v 0 -k 2. We rejected any exon target site for which there was more than a single match. Note that because any specific 13bp core sequence followed by the sequence NGG confers only 15bp of specificity, there should be on average ~5.6 matches to an extended core sequence in a random ~3Gb sequence (both strands). Therefore, most of the 1657793 initially identified targets were rejected; however 189864 sequences passed this filter. These comprise our set of CRISPR-targetable exonic locations in the human genome. The 189864 sequences target locations in 78028 merged exonic regions (~40.5% of the total of 192783 merged human exon regions) at a multiplicity of ~2.4 sites per targeted exonic region. To assess targeting at a gene level, we clustered RefSeq mRNA mappings so that any two RefSeq accessions (including the gene duplicates we distinguished in (ii)) that overlap a merged exon region are counted as a single gene cluster, the 189864 exonic

specific CRISPR sites target 17104 out of 18872 gene clusters (~90.6% of all gene clusters) at a multiplicity of ~11.1 per targeted gene cluster. (Note that while these gene clusters collapse RefSeq mRNA accessions that represent multiple isoforms of a single transcribed gene into a single entity, they will also collapse overlapping distinct genes as well as genes with antisense transcripts.) At the level of original RefSeq accessions, the 189864 sequences targeted exonic regions in 30563 out of a total of 43726 (~69.9%) mapped RefSeq accessions (including our distinguished gene duplicates) at a multiplicity of ~6.2 sites per targeted mapped RefSeq accession.

As we gather information on CRISPR performance at our computationally predicted human exon CRISPR target sites, we plan to refine our database by correlating performance with factors we expect to be important, such as base composition and secondary structure of both gRNAs and genomic targets (43, 44), and the epigenetic state of these targets in human cell lines for which this information is available (45).

Finally, we also incorporated these target sequences into a 200bp format that is compatible for multiplex synthesis on DNA arrays (14, 46). Our design allows for targeted retrieval of a specific or pools of gRNA sequences from the DNA array based oligonucleotide pool and its rapid cloning into a common expression vector (fig. S11A, Table S2). Specifically we tested this approach by synthesizing a 12k oligonucleotide pool from CustomArray Inc. (Table S3). Furthermore, as per our approach we were able to successfully retrieve gRNAs of choice from this library (fig. S11B). We observed an error rate of ~4 mutations per 1000bp of synthesized DNA.



B



U6 promoter + target RNA + guide RNA scaffold:

[illegible]

C

PAM
GNNNNNNN VNNNNNNNNNNNGG general form of target sequence

Name	Target Sequence
GFP gRNA Target 1	GTGAACCGCATCTGAGCTGAAGGG
GFP gRNA Target 2	GGAGCGCACCATCTTCTTCAAGG
AAVS1 gRNA Target 1	GTCCCCCTCCACCCACAGTGGGG
AAVS1 gRNA Target 2	GGGGCCACTAGGGACAGGATTGG
DNMT3a gRNA Target 1	GCATGATGCGCGGCCCAAGGAGG
DNMT3a gRNA Target 2	GAGATGATCGCCCCCTTCTTCTGG
DNMT3b gRNA Target	GAATTACTCAGCCCCAAGGAGG

Fig. S1. The engineered type II CRISPR system for human cells. **(A)** Expression format and full sequence of the cas9 gene insert. The RuvC-like and HNH motifs, and the C-terminus SV40 NLS

are respectively highlighted by blue, brown and orange colors. **(B)** U6 promoter based expression scheme for the guide RNAs and predicted RNA transcript secondary structure. The use of the U6 promoter constrains the 1st position in the RNA transcript to be a 'G' and thus all genomic sites of the form GN₂₀GG can be targeted using this approach. **(C)** The 7 gRNAs used in this study are listed.

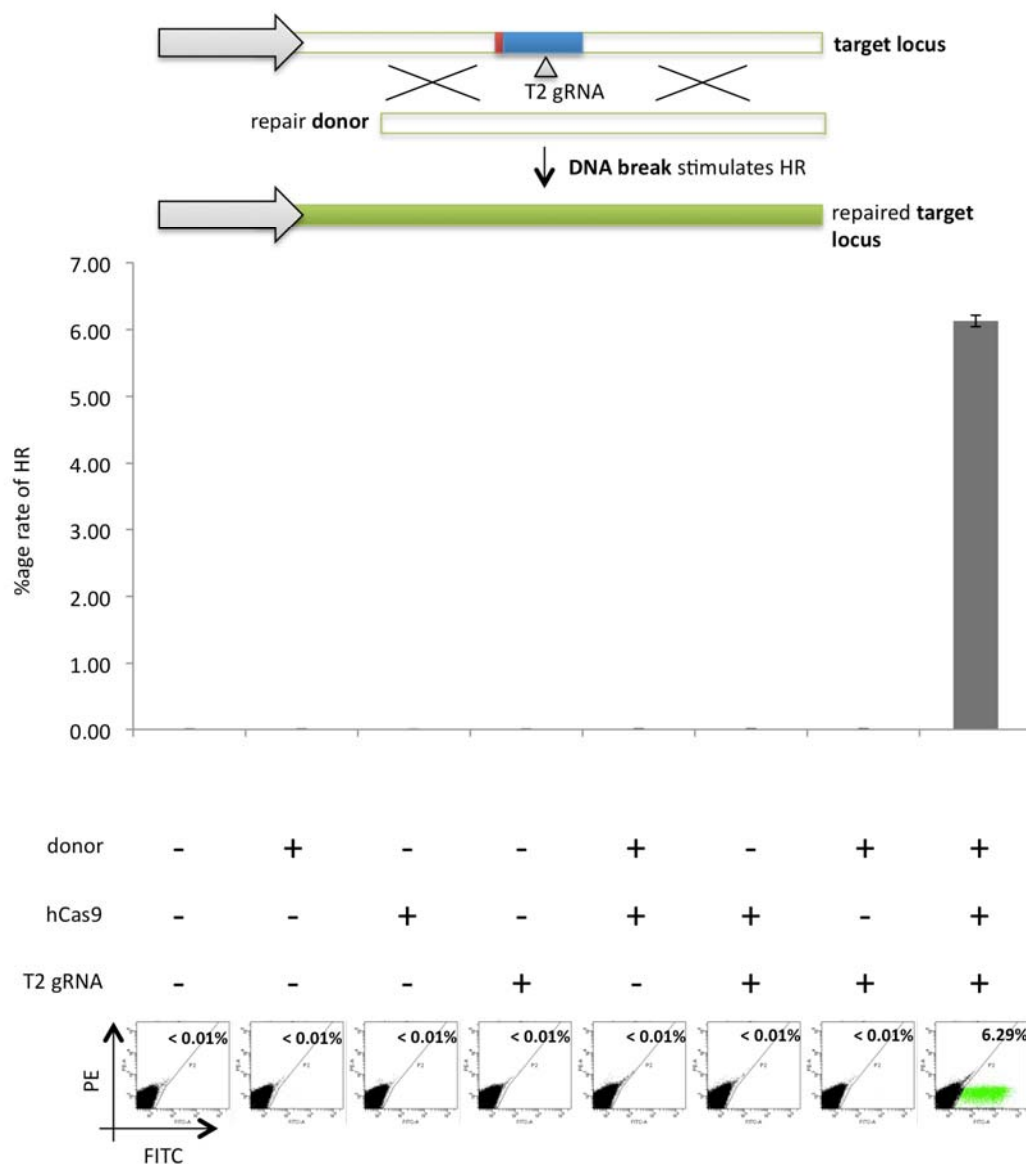


Fig. S2. RNA-guided genome editing requires both Cas9 and guide RNA for successful targeting. Using the GFP reporter assay described in Fig. 1B, all possible combinations of the repair DNA donor, Cas9 protein, and gRNA were tested for their ability to effect successful HR (in 293Ts). GFP+ cells were observed only when all the 3 components were present, validating that these CRISPR components are essential for RNA-guided genome editing. Data is mean +/- SEM (N=3).

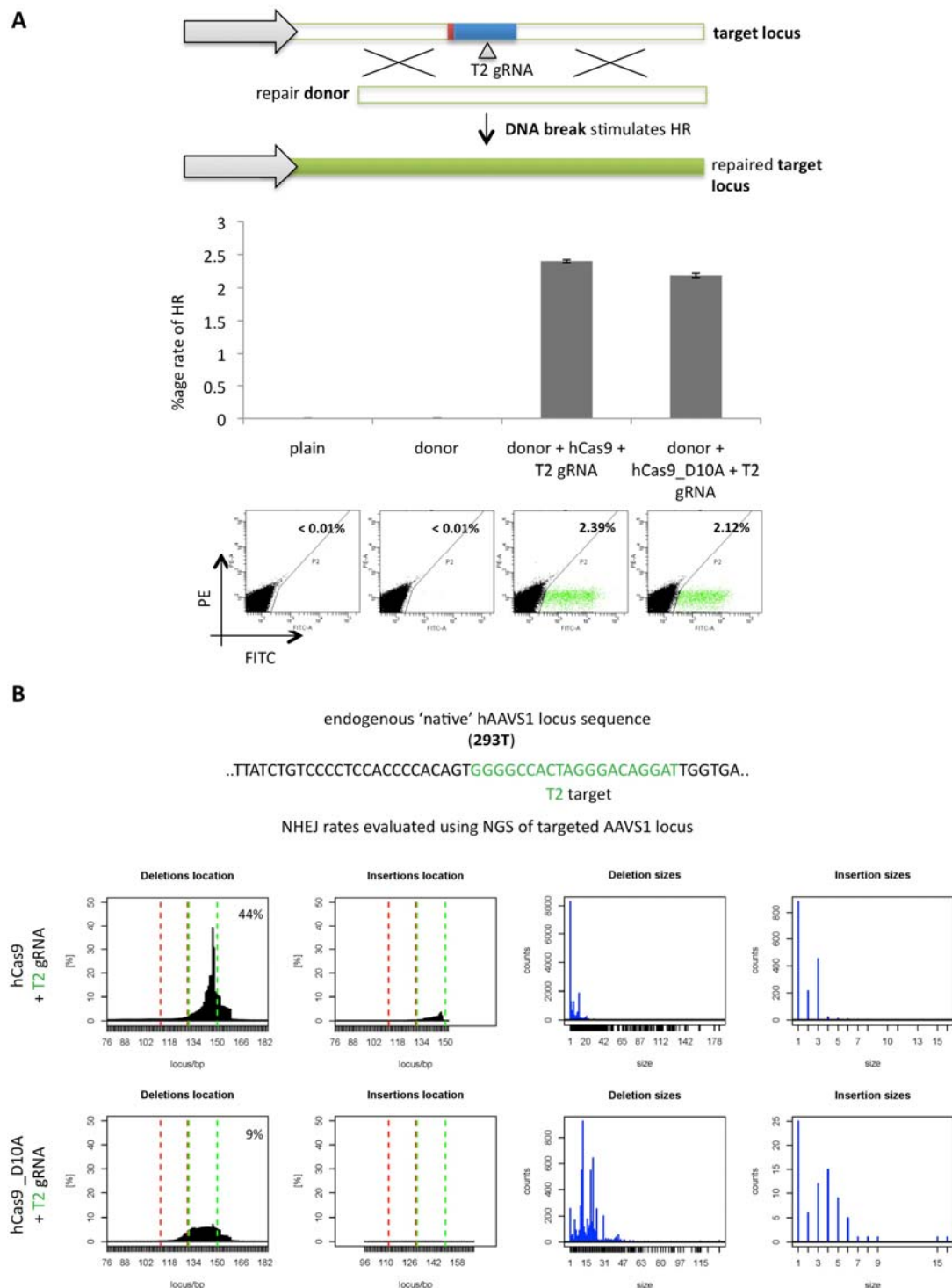


Fig. S3. Analysis of gRNA and Cas9 mediated genome editing. We closely examined the CRISPR mediated genome editing process using either **(A)** a GFP reporter assay as described earlier, and **(B)** deep sequencing of the targeted loci (in 293Ts). As comparison we also tested a D10A

mutant for Cas9 that has been shown in earlier reports to function as a nickase in *in vitro* assays. Our data shows that both Cas9 and Cas9D10A can effect successful HR at nearly similar rates. Deep sequencing however confirms that while Cas9 shows robust NHEJ at the targeted loci, the D10A mutant has significantly diminished NHEJ rates (as would be expected from its putative ability to only nick DNA). Also, consistent with the known biochemistry of the Cas9 protein, our NHEJ data confirms that most base-pair deletions or insertions occurred near the 3' end of the target sequence: the peak is ~3-4 bases upstream of the PAM site, with a median deletion frequency of ~9-10bp. Data is mean +/- SEM (N=3).

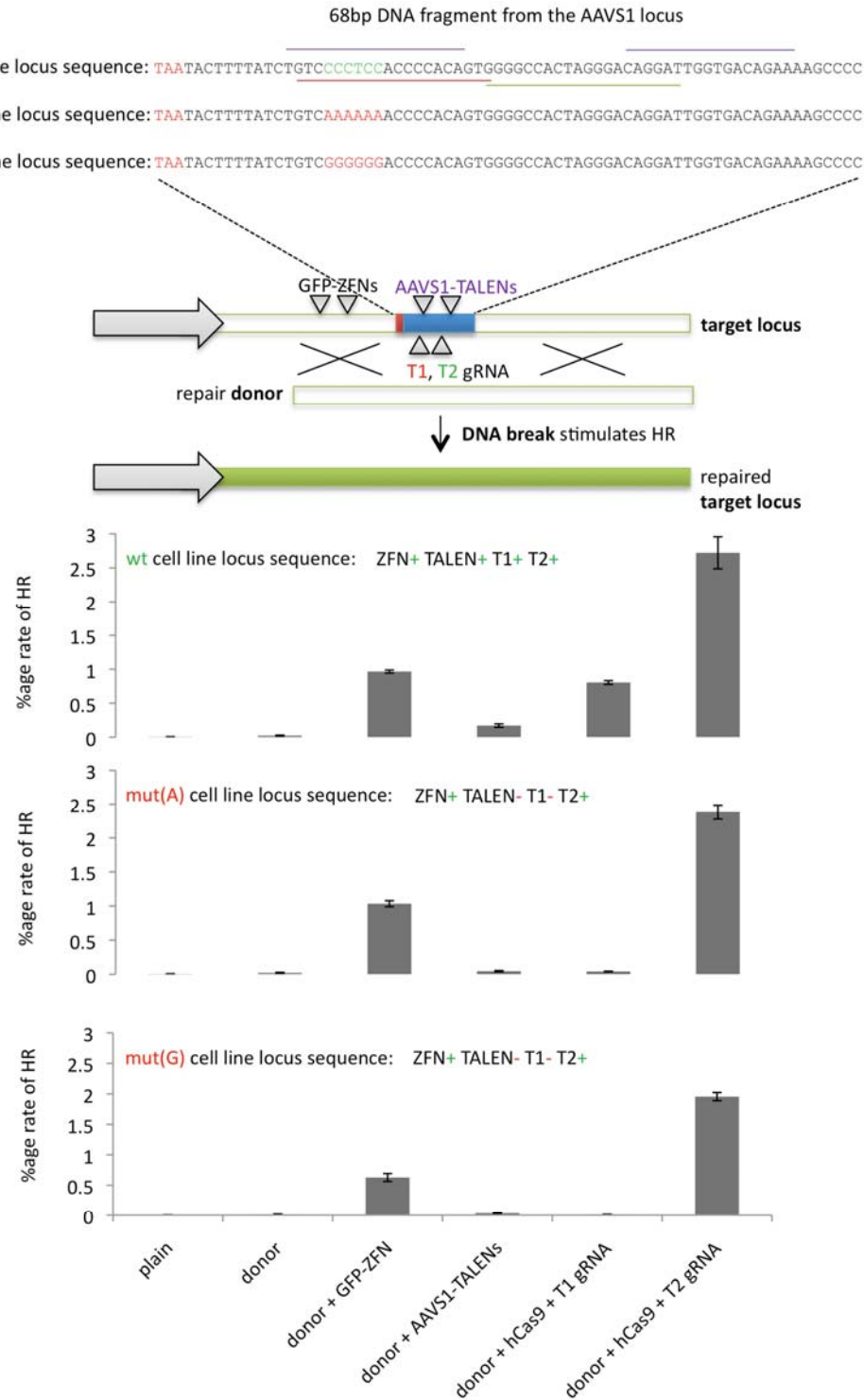


Fig. S4. RNA-guided genome editing is target sequence specific. Similar to the GFP reporter assay described in Fig. 1B, we developed 3 293T stable lines each bearing a distinct GFP reporter construct. These are distinguished by the sequence of the AAVS1 fragment insert (as

indicated in the figure). One line harbored the wild-type fragment while the two other lines were mutated at 6bp (highlighted in red). Each of the lines was then targeted by one of the following 4 reagents: a GFP-ZFN pair that can target all cell types since its targeted sequence was in the flanking GFP fragments and hence present in all cell lines; a AAVS1 TALEN that could potentially target only the wt-AAVS1 fragment since the mutations in the other two lines should render the left TALEN unable to bind their sites; the T1 gRNA which can also potentially target only the wt-AAVS1 fragment, since its target site is also disrupted in the two mutant lines; and finally the T2 gRNA which should be able to target all the 3 cell lines since unlike the T1 gRNA its target site is unaltered among the 3 lines. Consistent with these predictions, the ZFN modified all 3 cell types, the AAVS1 TALENs and the T1 gRNA only targeted the wt-AAVS1 cell type, and the T2 gRNA successfully targets all 3 cell types. These results together confirm that the guide RNA mediated editing is target sequence specific. Data is mean $\pm$ SEM (N=3).

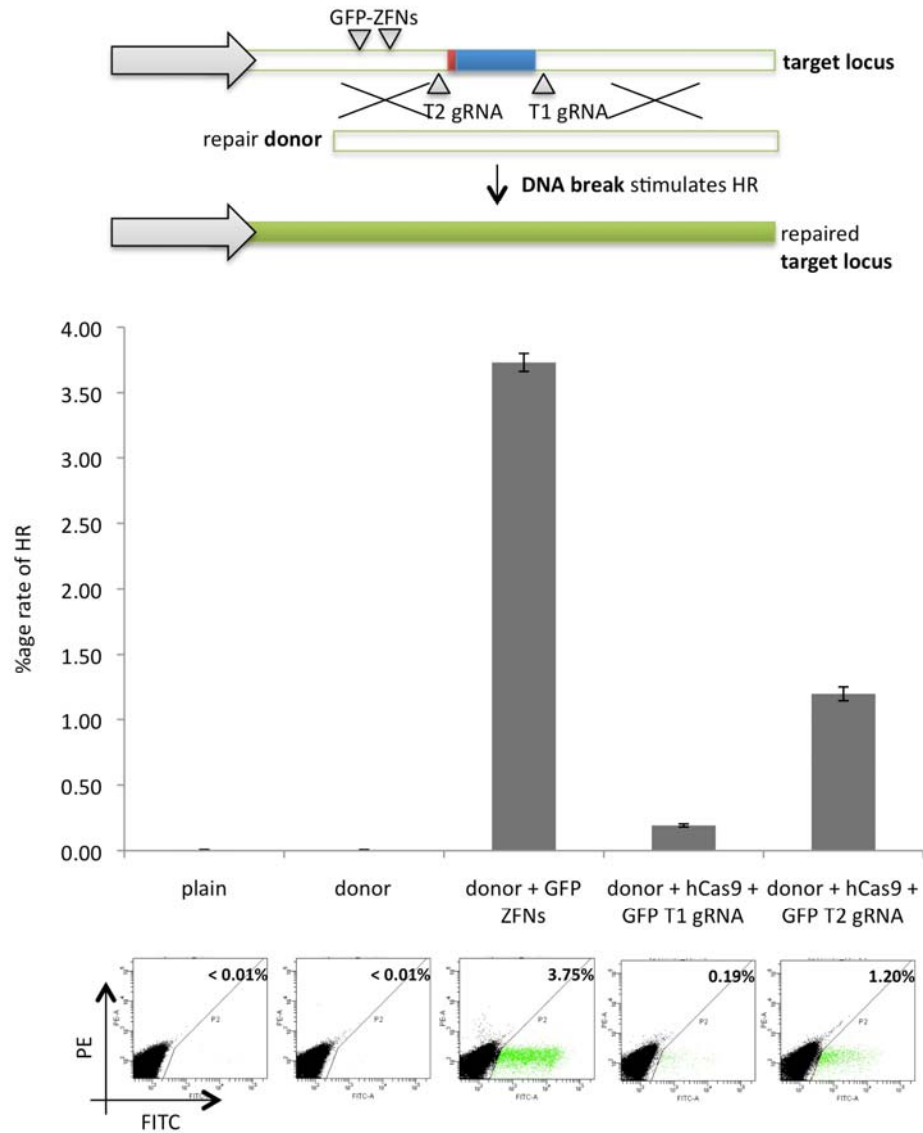


Fig. S5. Guide RNAs targeted to the GFP sequence enable robust genome editing. In addition to the 2 gRNAs targeting the AAVS1 insert, we also tested two additional gRNAs targeting the flanking GFP sequences of the reporter described in Fig. 1B (in 293Ts). These gRNAs were also able to effect robust HR at this engineered locus. Data is mean +/- SEM (N=3).

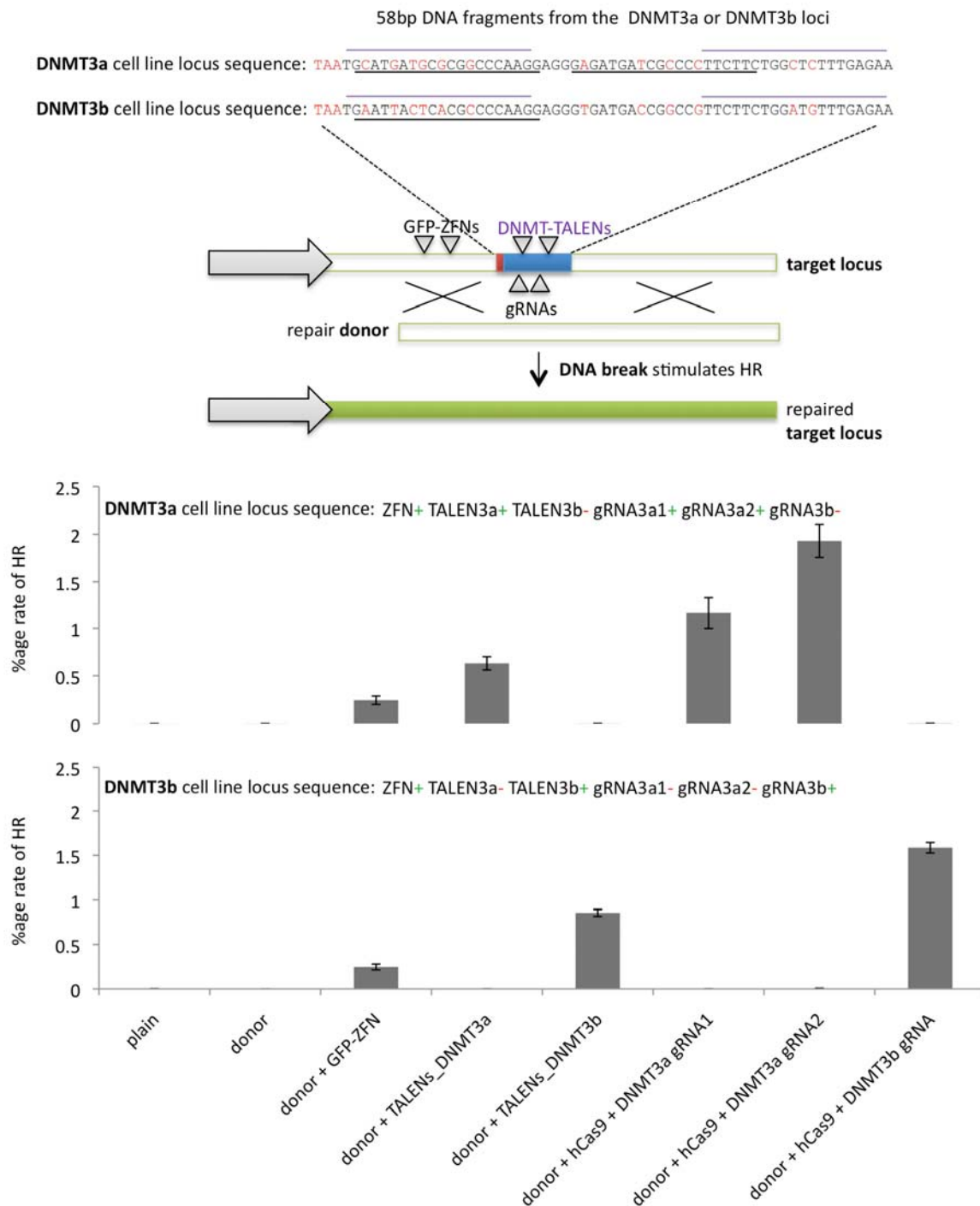


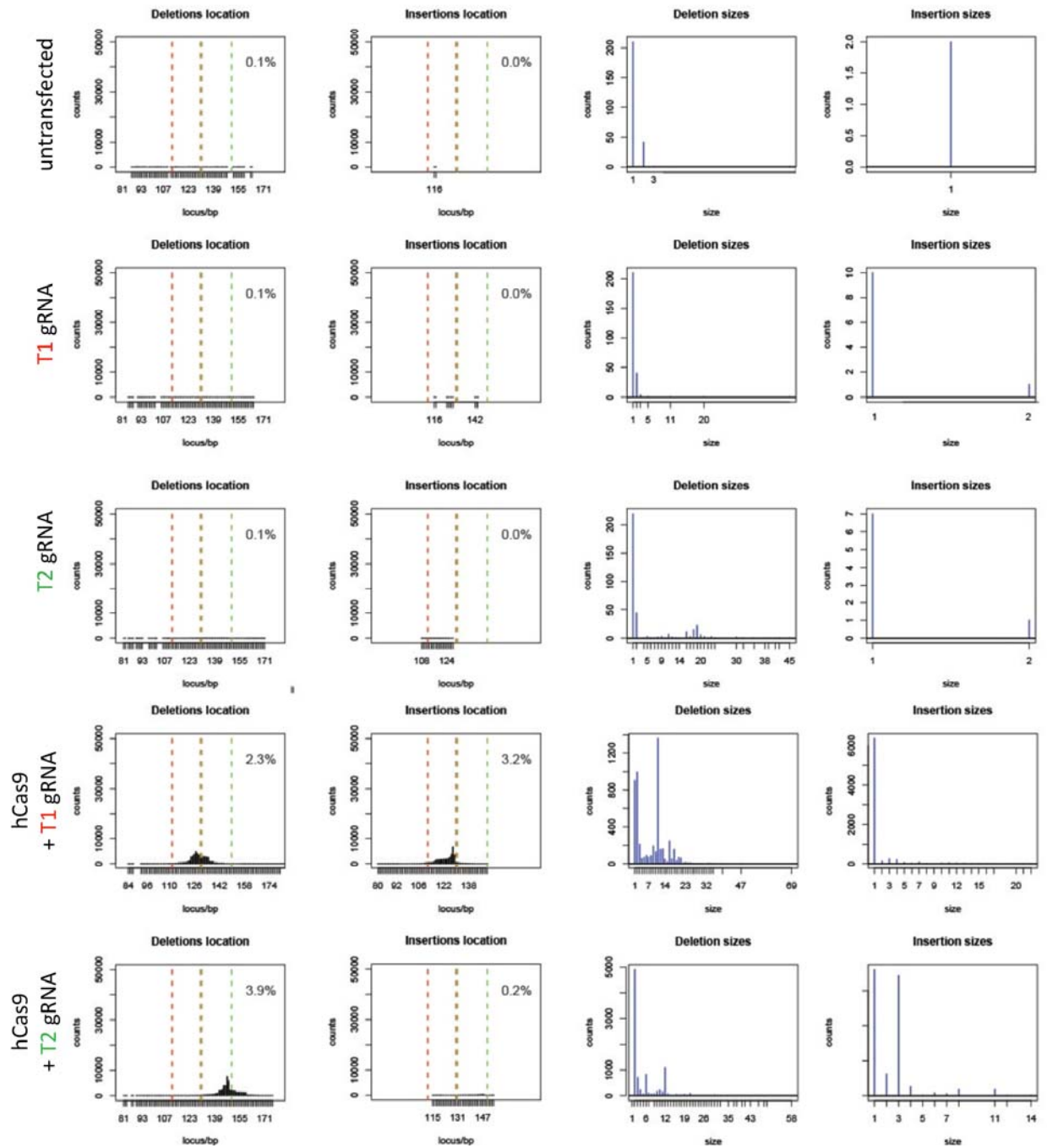
Fig. S6. RNA-guided genome editing is target sequence specific, and demonstrates similar targeting efficiencies as ZFNs or TALENs. Similar to the GFP reporter assay described in Fig. 1B, we developed 2 293T stable lines each bearing a distinct GFP reporter construct. These are distinguished by the sequence of the fragment insert (as indicated in the figure). One line

harbored a 58bp fragment from the DNMT3a gene while the other line bore a homologous 58bp fragment from the DNMT3b gene. The sequence differences are highlighted in red. Each of the lines was then targeted by one of the following 6 reagents: a GFP-ZFN pair that can target all cell types since its targeted sequence was in the flanking GFP fragments and hence present in all cell lines; a pair of TALENs that potentially target either DNMT3a or DNMT3b fragments; a pair of gRNAs that can potentially target only the DNMT3a fragment; and finally a gRNA that should potentially only target the DNMT3b fragment. Consistent with these predictions, the ZFN modified all 3 cell types, and the TALENs and gRNAs only their respective targets. Furthermore the efficiencies of targeting were comparable across the 6 targeting reagents. These results together confirm that RNA-guided editing is target sequence specific and demonstrates similar targeting efficiencies as ZFNs or TALENs. Data is mean $\pm$ SEM (N=3).

endogenous 'native' hAAVS1 locus sequence
(PGP1 iPS)

..TTATCTG**TCCCTCCACCCACAGT**GGGCCACTAGGGACAGGATTGGTGA..
T1 target T2 target

NHEJ rates evaluated using NGS of targeted AAVS1 locus



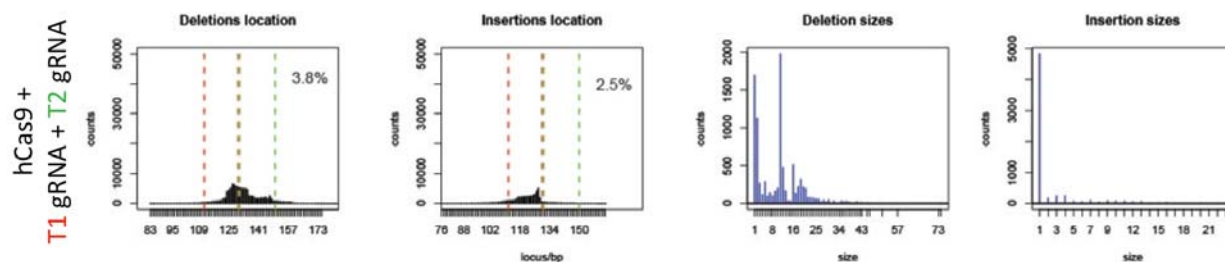


Fig. S7. RNA-guided NHEJ in human iPS cells. We nucleofected human iPS cells (PGP1) with constructs indicated in the left panel. 4 days after nucleofection, we measured NHEJ rate by assessing genomic deletion and insertion rate at double-strand breaks (DSBs) by deep sequencing. *Panel 1:* Deletion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site; green dash lines: boundary of T2 RNA targeting site. We plot the deletion incidence at each nucleotide position in black lines and we calculated the deletion rate as the percentage of reads carrying deletions. *Panel 2:* Insertion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site; green dash lines: boundary of T2 RNA targeting site. We plot the incidence of insertion at the genomic location where the first insertion junction was detected in black lines and we calculated the insertion rate as the percentage of reads carrying insertions. *Panel 3:* Deletion size distribution. We plot the frequencies of different size deletions among the whole NHEJ population. *Panel 4:* insertion size distribution. We plot the frequencies of different sizes insertions among the whole NHEJ population. iPS targeting by both gRNAs is efficient (2-4%), sequence specific (as shown by the shift in position of the NHEJ deletion distributions), and reaffirming the results of fig. S2, the NGS-based analysis also shows that both the Cas9 protein and the gRNA are essential for NHEJ events at the target locus.

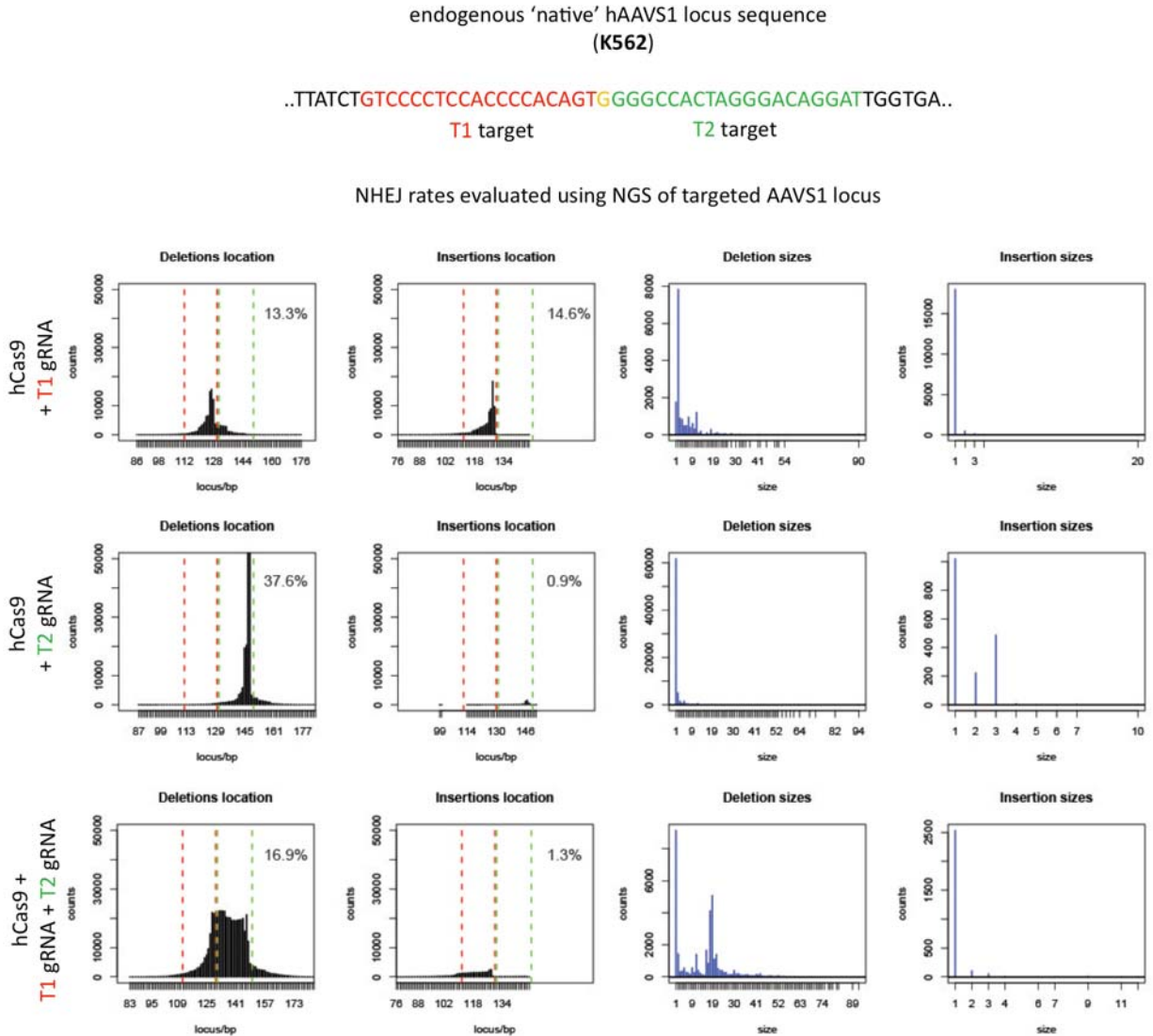


Fig. S8. RNA-guided NHEJ in K562 cells. We nucleofected K562 cells with constructs indicated in the left panel. 4 days after nucleofection, we measured NHEJ rate by assessing genomic deletion and insertion rate at DSBs by deep sequencing. *Panel 1:* Deletion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site; green dash lines: boundary of T2 RNA targeting site. We plot the deletion incidence at each nucleotide position in black lines and calculated the deletion rate as the percentage of reads carrying deletions. *Panel 2:* Insertion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site;

green dash lines: boundary of T2 RNA targeting site. We plot the incidence of insertion at the genomic location where the first insertion junction was detected in black lines and we calculated the insertion rate as the percentage of reads carrying insertions. *Panel 3*: Deletion size distribution. We plot the frequencies of different size deletions among the whole NHEJ population. *Panel 4*: insertion size distribution. We plot the frequencies of different sizes insertions among the whole NHEJ population. K562 targeting by both gRNAs is efficient (13-38%) and sequence specific (as shown by the shift in position of the NHEJ deletion distributions). Importantly, as evidenced by the peaks in the histogram of observed frequencies of deletion sizes, simultaneous introduction of both T1 and T2 guide RNAs resulted in high efficiency deletion of the intervening 19bp fragment, demonstrating that multiplexed editing of genomic loci is also feasible using this approach.

endogenous 'native' hAAVS1 locus sequence
(293T)

..TTATCTGTCCCTCCACCCACAGTGGGCCACTAGGGACAGGATTGGTGA..
T1 target T2 target

NHEJ rates evaluated using NGS of targeted AAVS1 locus

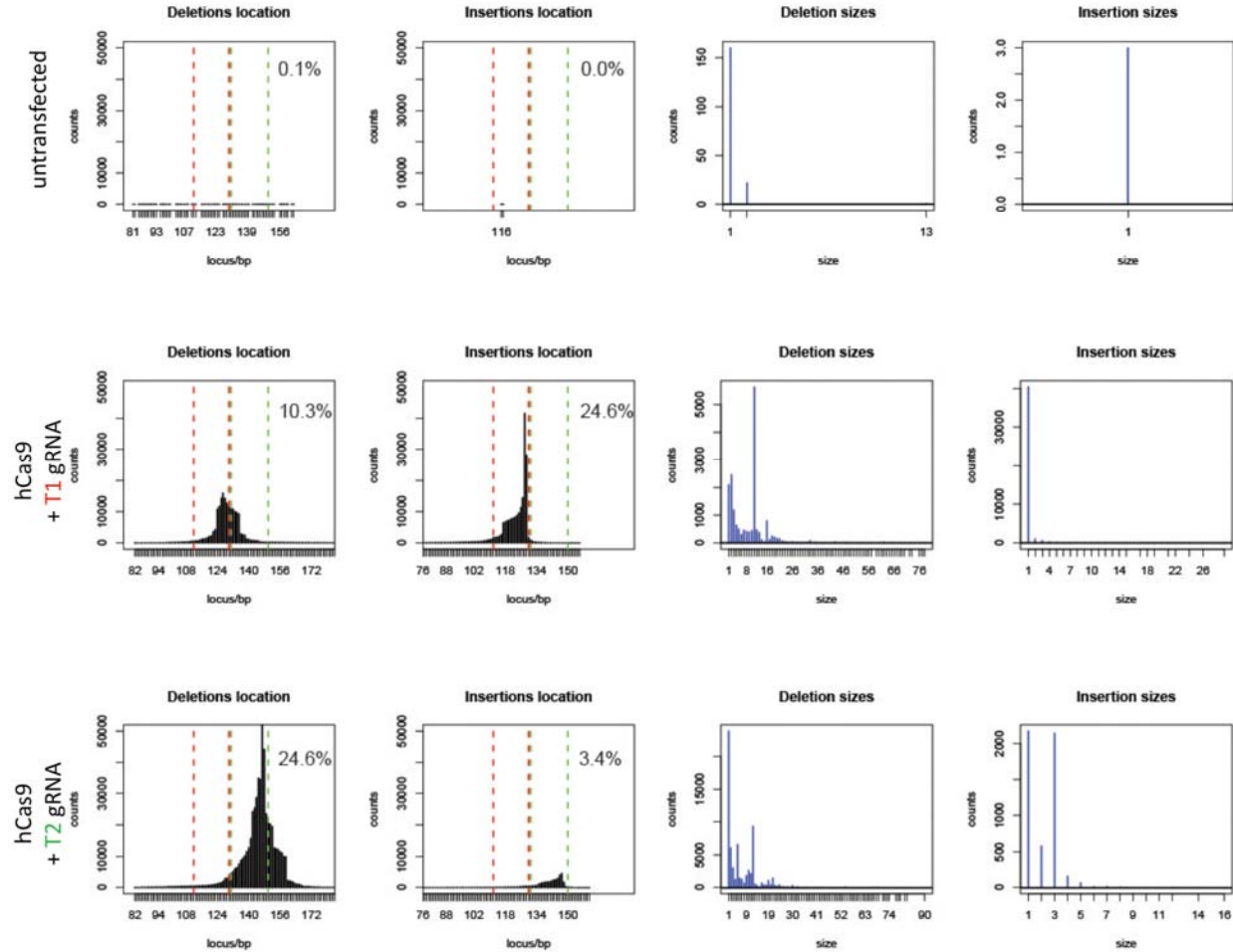


Fig. S9. RNA-guided NHEJ in 293T cells. We transfected 293T cells with constructs indicated in the left panel. 4 days after nucleofection, we measured NHEJ rate by assessing genomic deletion and insertion rate at DSBs by deep sequencing. *Panel 1:* Deletion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site; green dash lines: boundary of T2 RNA targeting site. We plot the deletion incidence at each nucleotide position in black lines and calculated the deletion rate as the percentage of reads carrying deletions. *Panel 2:*

Insertion rate detected at targeting region. Red dash lines: boundary of T1 RNA targeting site; green dash lines: boundary of T2 RNA targeting site. We plot the incidence of insertion at the genomic location where the first insertion junction was detected in black lines and we calculated the insertion rate as the percentage of reads carrying insertions. *Panel 3*: Deletion size distribution. We plot the frequencies of different size deletions among the whole NHEJ population. *Panel 4*: insertion size distribution. We plot the frequencies of different sizes insertions among the whole NHEJ population. 293T targeting by both gRNAs is efficient (10-24%) and sequence specific (as shown by the shift in position of the NHEJ deletion distributions).

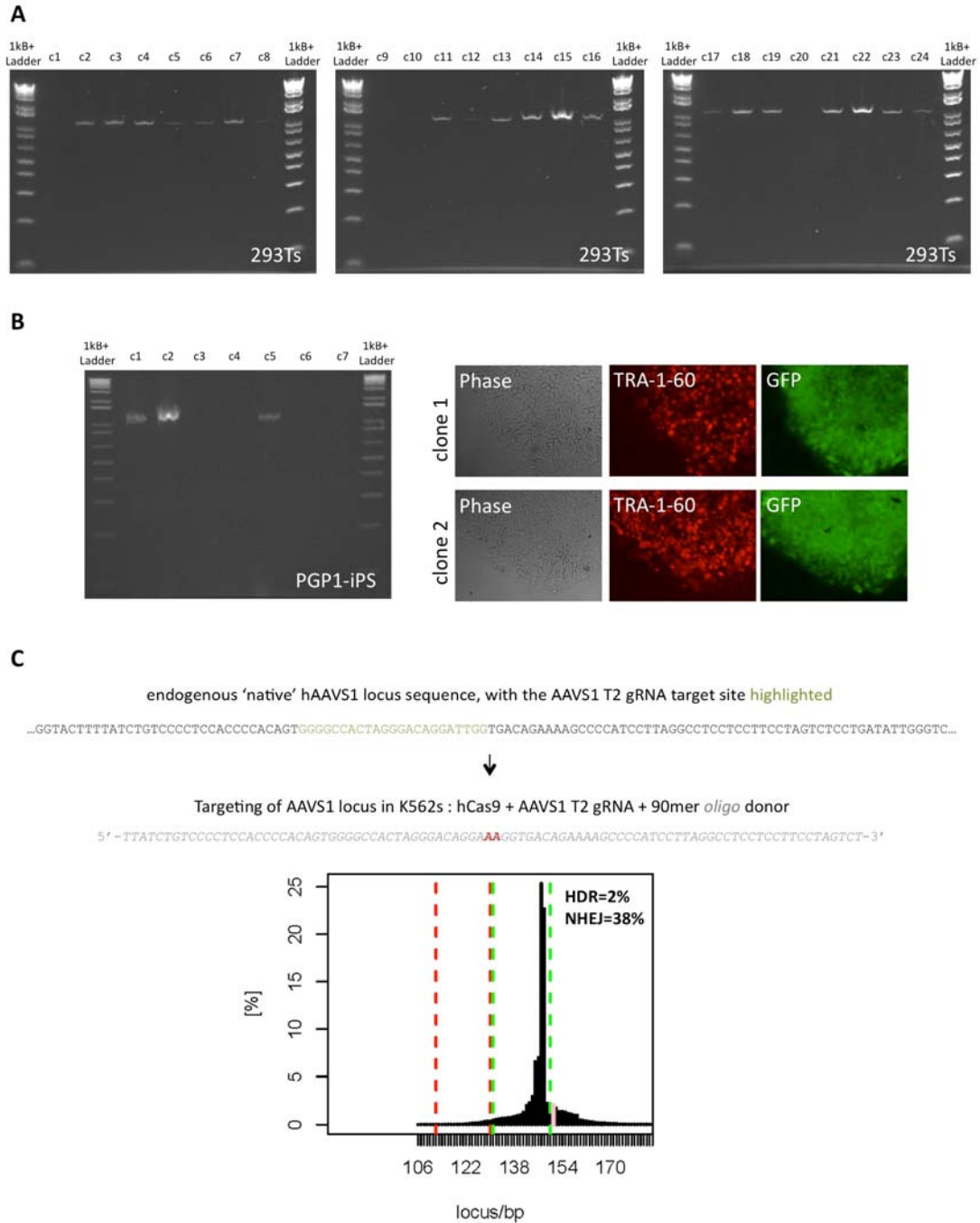


Fig. S10. HR at the endogenous AAVS1 locus using either a dsDNA donor or a short oligonucleotide donor. **(A)** PCR screen (refer Fig. 2C) confirmed that 21/24 randomly picked 293T clones were successfully targeted. **(B)** Similar PCR screen confirmed 3/7 randomly picked

PGP1-iPS clones were also successfully targeted. (C) Finally short 90mer oligos could also effect robust targeting at the endogenous AAVS1 locus (shown here for K562 cells).

(i) targeted retrieval of a specific or pools of gRNA targets from the DNA array oligonucleotide pool (through 3 sequential rounds of nested PCR as indicated in the figure schematic); and (ii) its rapid cloning into a common expression vector which upon linearization using an AflIII site serves as a recipient for Gibson assembly mediated incorporation of the gRNA insert fragment.

(B) We confirmed this methodology by targeted retrieval of 10 unique gRNAs from a 12k oligonucleotide pool synthesized by CustomArray Inc. (refer Methods).

References

1. B. Wiedenheft, S. H. Sternberg, J. A. Doudna, RNA-guided genetic silencing systems in bacteria and archaea. *Nature* **482**, 331 (2012). [doi:10.1038/nature10886](https://doi.org/10.1038/nature10886) [Medline](#)
2. D. Bhaya, M. Davison, R. Barrangou, CRISPR-Cas systems in bacteria and archaea: Versatile small RNAs for adaptive defense and regulation. *Annu. Rev. Genet.* **45**, 273 (2011). [doi:10.1146/annurev-genet-110410-132430](https://doi.org/10.1146/annurev-genet-110410-132430) [Medline](#)
3. M. P. Terns, R. M. Terns, CRISPR-based adaptive immune systems. *Curr. Opin. Microbiol.* **14**, 321 (2011). [doi:10.1016/j.mib.2011.03.005](https://doi.org/10.1016/j.mib.2011.03.005) [Medline](#)
4. M. Jinek *et al.*, A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science* **337**, 816 (2012). [doi:10.1126/science.1225829](https://doi.org/10.1126/science.1225829) [Medline](#)
5. G. Gasiunas, R. Barrangou, P. Horvath, V. Siksnys, Cas9-crRNA ribonucleoprotein complex mediates specific DNA cleavage for adaptive immunity in bacteria. *Proc. Natl. Acad. Sci. U.S.A.* **109**, E2579 (2012). [doi:10.1073/pnas.1208507109](https://doi.org/10.1073/pnas.1208507109) [Medline](#)
6. R. Sapranauskas *et al.*, The *Streptococcus thermophilus* CRISPR/Cas system provides immunity in *Escherichia coli*. *Nucleic Acids Res.* **39**, 9275 (2011). [doi:10.1093/nar/gkr606](https://doi.org/10.1093/nar/gkr606) [Medline](#)
7. T. R. Brummelkamp, R. Bernards, R. Agami, A system for stable expression of short interfering RNAs in mammalian cells. *Science* **296**, 550 (2002). [doi:10.1126/science.1068999](https://doi.org/10.1126/science.1068999) [Medline](#)
8. M. Miyagishi, K. Taira, U6 promoter-driven siRNAs with four uridine 3' overhangs efficiently suppress targeted gene expression in mammalian cells. *Nat. Biotechnol.* **20**, 497 (2002). [doi:10.1038/nbt0502-497](https://doi.org/10.1038/nbt0502-497) [Medline](#)
9. E. Deltcheva *et al.*, CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III. *Nature* **471**, 602 (2011). [doi:10.1038/nature09886](https://doi.org/10.1038/nature09886) [Medline](#)
10. J. Zou, P. Mali, X. Huang, S. N. Dowey, L. Cheng, Site-specific gene correction of a point mutation in human iPS cells derived from an adult patient with sickle cell disease. *Blood* **118**, 4599 (2011). [doi:10.1182/blood-2011-02-335554](https://doi.org/10.1182/blood-2011-02-335554) [Medline](#)
11. N. E. Sanjana *et al.*, A transcription activator-like effector toolbox for genome engineering. *Nat. Protoc.* **7**, 171 (2012). [doi:10.1038/nprot.2011.431](https://doi.org/10.1038/nprot.2011.431) [Medline](#)
12. J. H. Lee *et al.*, A robust approach to identifying tissue-specific gene expression regulatory variants using personalized human induced pluripotent stem cells. *PLoS Genet.* **5**, e1000718 (2009). [doi:10.1371/journal.pgen.1000718](https://doi.org/10.1371/journal.pgen.1000718) [Medline](#)
13. D. Hockemeyer *et al.*, Efficient targeting of expressed and silent genes in human ESCs and iPSCs using zinc-finger nucleases. *Nat. Biotechnol.* **27**, 851 (2009). [doi:10.1038/nbt.1562](https://doi.org/10.1038/nbt.1562) [Medline](#)
14. S. Kosuri *et al.*, Scalable gene synthesis by selective amplification of DNA pools from high-fidelity microchips. *Nat. Biotechnol.* **28**, 1295 (2010). [doi:10.1038/nbt.1716](https://doi.org/10.1038/nbt.1716) [Medline](#)
15. V. Pattanayak, C. L. Ramirez, J. K. Joung, D. R. Liu, Revealing off-target cleavage specificities of zinc-finger nucleases by in vitro selection. *Nat. Methods* **8**, 765 (2011). [doi:10.1038/nmeth.1670](https://doi.org/10.1038/nmeth.1670) [Medline](#)

16. N. M. King, O. Cohen-Haguenuer, En route to ethical recommendations for gene transfer clinical trials. *Mol. Ther.* **16**, 432 (2008). [doi:10.1038/mt.2008.13](https://doi.org/10.1038/mt.2008.13) [Medline](#)
17. Y. G. Kim, J. Cha, S. Chandrasegaran, Hybrid restriction enzymes: Zinc finger fusions to Fok I cleavage domain. *Proc. Natl. Acad. Sci. U.S.A.* **93**, 1156 (1996). [doi:10.1073/pnas.93.3.1156](https://doi.org/10.1073/pnas.93.3.1156) [Medline](#)
18. E. J. Rebar, C. O. Pabo, Zinc finger phage: Affinity selection of fingers with new DNA-binding specificities. *Science* **263**, 671 (1994). [doi:10.1126/science.8303274](https://doi.org/10.1126/science.8303274) [Medline](#)
19. J. Boch *et al.*, Breaking the code of DNA binding specificity of TAL-type III effectors. *Science* **326**, 1509 (2009). [doi:10.1126/science.1178811](https://doi.org/10.1126/science.1178811) [Medline](#)
20. M. J. Moscou, A. J. Bogdanove, A simple cipher governs DNA recognition by TAL effectors. *Science* **326**, 1501 (2009). [doi:10.1126/science.1178817](https://doi.org/10.1126/science.1178817) [Medline](#)
21. L. Cong *et al.*, Multiplex genome engineering using CRISPR/Cas systems. *Science* **326**, 1231143 (2013). [doi:10.1126/science.1231143](https://doi.org/10.1126/science.1231143)
22. A. S. Khalil, J. J. Collins, Synthetic biology: Applications come of age. *Nat. Rev. Genet.* **11**, 367 (2010). [doi:10.1038/nrg2775](https://doi.org/10.1038/nrg2775) [Medline](#)
23. P. E. Purnick, R. Weiss, The second wave of synthetic biology: From modules to systems. *Nat. Rev. Mol. Cell Biol.* **10**, 410 (2009). [doi:10.1038/nrm2698](https://doi.org/10.1038/nrm2698) [Medline](#)
24. J. Zou *et al.*, Gene targeting of a disease-related gene in human induced pluripotent stem and embryonic stem cells. *Cell Stem Cell* **5**, 97 (2009). [doi:10.1016/j.stem.2009.05.023](https://doi.org/10.1016/j.stem.2009.05.023) [Medline](#)
25. N. Holt *et al.*, Human hematopoietic stem/progenitor cells modified by zinc-finger nucleases targeted to CCR5 control HIV-1 in vivo. *Nat. Biotechnol.* **28**, 839 (2010). [doi:10.1038/nbt.1663](https://doi.org/10.1038/nbt.1663) [Medline](#)
26. F. D. Urnov *et al.*, Highly efficient endogenous human gene correction using designed zinc-finger nucleases. *Nature* **435**, 646 (2005). [doi:10.1038/nature03556](https://doi.org/10.1038/nature03556) [Medline](#)
27. A. Lombardo *et al.*, Gene editing in human stem cells using zinc finger nucleases and integrase-defective lentiviral vector delivery. *Nat. Biotechnol.* **25**, 1298 (2007). [doi:10.1038/nbt1353](https://doi.org/10.1038/nbt1353) [Medline](#)
28. H. Li *et al.*, In vivo genome editing restores haemostasis in a mouse model of haemophilia. *Nature* **475**, 217 (2011). [doi:10.1038/nature10177](https://doi.org/10.1038/nature10177) [Medline](#)
29. K. S. Makarova *et al.*, Evolution and classification of the CRISPR-Cas systems. *Nat. Rev. Microbiol.* **9**, 467 (2011). [doi:10.1038/nrmicro2577](https://doi.org/10.1038/nrmicro2577) [Medline](#)
30. P. Horvath, R. Barrangou, CRISPR/Cas, the immune system of bacteria and archaea. *Science* **327**, 167 (2010). [doi:10.1126/science.1179555](https://doi.org/10.1126/science.1179555) [Medline](#)
31. H. Deveau *et al.*, Phage response to CRISPR-encoded resistance in *Streptococcus thermophilus*. *J. Bacteriol.* **190**, 1390 (2008). [doi:10.1128/JB.01412-07](https://doi.org/10.1128/JB.01412-07) [Medline](#)
32. J. R. van der Ploeg, Analysis of CRISPR in *Streptococcus mutans* suggests frequent occurrence of acquired immunity against infection by M102-like bacteriophages. *Microbiology* **155**, 1966 (2009). [doi:10.1099/mic.0.027508-0](https://doi.org/10.1099/mic.0.027508-0) [Medline](#)

33. M. Rho, Y. W. Wu, H. Tang, T. G. Doak, Y. Ye, Diverse CRISPRs evolving in human microbiomes. *PLoS Genet.* **8**, e1002441 (2012). [doi:10.1371/journal.pgen.1002441](https://doi.org/10.1371/journal.pgen.1002441) [Medline](#)
34. D. T. Pride *et al.*, Analysis of streptococcal CRISPRs from human saliva reveals substantial sequence diversity within and between subjects over time. *Genome Res.* **21**, 126 (2011). [doi:10.1101/gr.111732.110](https://doi.org/10.1101/gr.111732.110) [Medline](#)
35. J. E. Garneau *et al.*, The CRISPR/Cas bacterial immune system cleaves bacteriophage and plasmid DNA. *Nature* **468**, 67 (2010). [doi:10.1038/nature09523](https://doi.org/10.1038/nature09523) [Medline](#)
36. K. M. Esvelt, J. C. Carlson, D. R. Liu, A system for the continuous directed evolution of biomolecules. *Nature* **472**, 499 (2011). [doi:10.1038/nature09929](https://doi.org/10.1038/nature09929) [Medline](#)
37. R. Barrangou, P. Horvath, CRISPR: New horizons in phage resistance and strain identification. *Annu. Rev. Food Sci. Technol.* **3**, 143 (2012). [doi:10.1146/annurev-food-022811-101134](https://doi.org/10.1146/annurev-food-022811-101134) [Medline](#)
38. W. J. Kent *et al.*, The human genome browser at UCSC. *Genome Res.* **12**, 996 (2002). [Medline](#)
39. T. R. Dreszer *et al.*, The UCSC Genome Browser database: Extensions and updates 2011. *Nucleic Acids Res.* **40**, (Database issue), D918 (2012). [doi:10.1093/nar/gkr1055](https://doi.org/10.1093/nar/gkr1055) [Medline](#)
40. D. Karolchik *et al.*, The UCSC Table Browser data retrieval tool. *Nucleic Acids Res.* **32**, (Database issue), D493 (2004). [doi:10.1093/nar/gkh103](https://doi.org/10.1093/nar/gkh103) [Medline](#)
41. A. R. Quinlan, I. M. Hall, BEDTools: A flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841 (2010). [doi:10.1093/bioinformatics/btq033](https://doi.org/10.1093/bioinformatics/btq033) [Medline](#)
42. B. Langmead, C. Trapnell, M. Pop, S. L. Salzberg, Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol.* **10**, R25 (2009). [doi:10.1186/gb-2009-10-3-r25](https://doi.org/10.1186/gb-2009-10-3-r25) [Medline](#)
43. R. Lorenz *et al.*, ViennaRNA Package 2.0. *Algorithms Mol. Biol.* **6**, 26 (2011). [doi:10.1186/1748-7188-6-26](https://doi.org/10.1186/1748-7188-6-26) [Medline](#)
44. D. H. Mathews, J. Sabina, M. Zuker, D. H. Turner, Expanded sequence dependence of thermodynamic parameters improves prediction of RNA secondary structure. *J. Mol. Biol.* **288**, 911 (1999). [doi:10.1006/jmbi.1999.2700](https://doi.org/10.1006/jmbi.1999.2700) [Medline](#)
45. R. E. Thurman *et al.*, The accessible chromatin landscape of the human genome. *Nature* **489**, 75 (2012). [doi:10.1038/nature11232](https://doi.org/10.1038/nature11232) [Medline](#)
46. Q. Xu, M. R. Schlach, G. J. Hannon, S. J. Elledge, Design of 240,000 orthogonal 25mer DNA barcode probes. *Proc. Natl. Acad. Sci. U.S.A.* **106**, 2289 (2009). [doi:10.1073/pnas.0812506106](https://doi.org/10.1073/pnas.0812506106) [Medline](#)